GATE

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EC : ELECTRONICS AND COMMUNICATIONS

Solutions on Topicwise Examination Questions

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2000

1. (D)

$$V(t) = \frac{Ldi}{dt} = \frac{d}{dt}(e^{at} + e^{bt}) = ae^{at} + be^{bt}$$

2. (C)

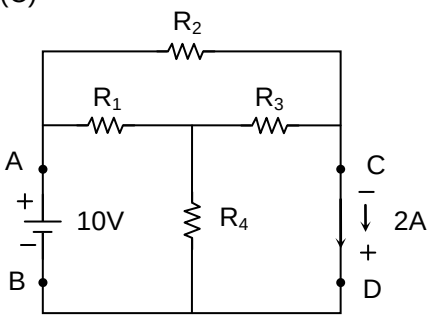


Fig. (a)

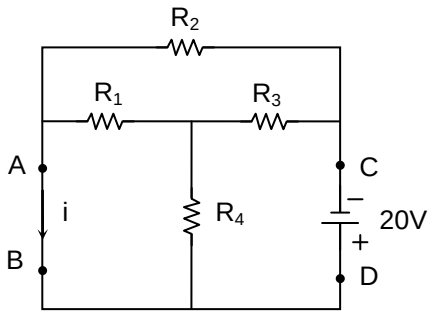


Fig. (b)

From figure A,
10V battery is applied between points A and B and current is measured between C and D.

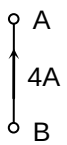
A $\rightarrow$ higher potential	C $\rightarrow$ higher potential
B $\rightarrow$ lower potential	D $\rightarrow$ lower potential
For voltage source	For current measured

now, we apply reciprocity theorem.
now, 20V battery is connected between C and D with D at higher potential and C at lower potential.

$\therefore$ Current in AB will be reversed.
i.e. A at lower potential
B at higher potential

$\therefore$ Current flow through AB branch will be

$\therefore$ By superposition theorem
For 20V battery between C and D with polarity as shown in figure, current through AB will be



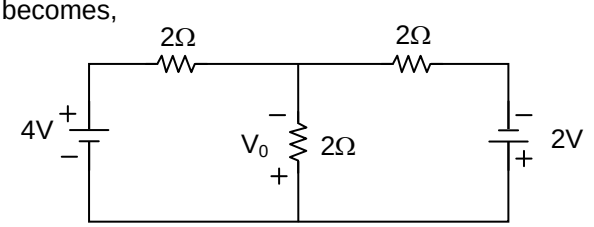
$\therefore$ i = – 4A
which is opposite to the direction of i.

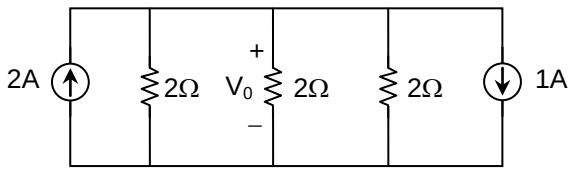
3. (D)

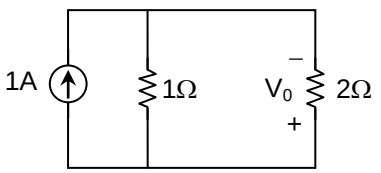
Diode will be forward biased.

Since, cut-in voltage and forward resistance of the diode are not given, we assume the diode to be ideal.

Circuit becomes,







By current divider rule,

$$i_{2\Omega} = \frac{1\times 1}{(1+2)} = \frac{1}{3}\text{ A}$$

$$\therefore \quad V_o = -2 \times \frac{1}{3} = \frac{-2}{3}\text{ V}$$

4. (A)

For d.c. voltage capacitor acts as open circuit

$\therefore$ 3V appears as it is.

$$i = \frac{V_m \sin(\omega t - \phi)}{Z}$$

$$V_0 = i \,.\, X_C = \frac{V_m \sin(\omega t - \phi)}{Z} X_C$$

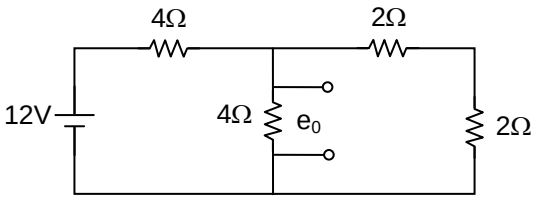
$$X_C = \frac{1}{2\pi fC} = -1000\ ;$$

$$Z = \sqrt{R^2 + X_C{}^2} = 1000\ ; \qquad \phi = \tan^{-1}\frac{X_C}{R} = \frac{\pi}{4}$$

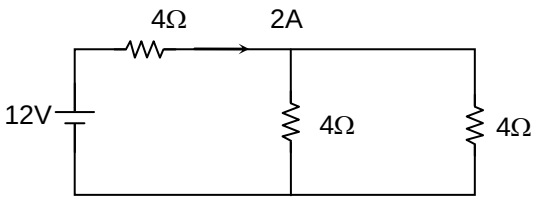
Ans. $3 + \frac{4}{\sqrt{2}}\sin(100t - \pi/4)$ V

2001

1. (C)



Simplifying the circuit, we get



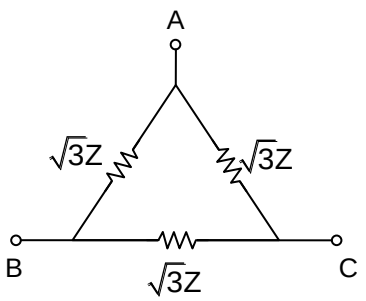
$i_{4\Omega} = 1A$

$\therefore \quad e_0 = 4 \times 1$

$e_0 = 4V$

2. (A)

Delta circuit

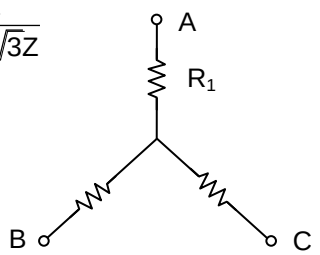


Wye circuit

$$R_1 = \frac{\sqrt{3}Z \times \sqrt{3}Z}{\sqrt{3}Z + \sqrt{3}Z + \sqrt{3}Z}$$

$$= \frac{3Z^2}{3\sqrt{3}\,Z}$$

$$= \frac{Z}{\sqrt{3}}$$



3. (A)

Maximum power will be delivered when,

$$R = \sqrt{R_L^{\,2} + (\omega L)^2}$$

$\omega = 10$, $R_L = 10\Omega$, $L = 1H$

$$R = \sqrt{10^2 + 10^2} = 14.14\ \Omega$$

4. (A)

Figure (a)

$$I_2(\omega) = \frac{E_m \cos\omega t}{R_2 + \frac{1}{j\omega C}}$$

$$= E_m \cos\omega t \left\{ \frac{R_2}{R_2^{\,2} + \left(\frac{1}{\omega C}\right)^2} - \frac{\frac{1}{j\omega C}}{R_2^{\,2} + \left(\frac{1}{\omega C}\right)^2} \right\}$$

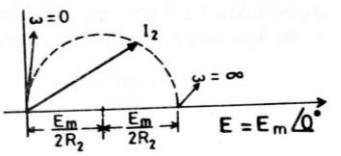


Fig. (a)

At $\omega = \infty$, $I_2(\omega) = \frac{E_m}{R_2}$

Also $\omega = 0$, $I_2(0) = 0$

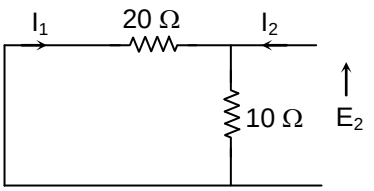
Thus the derived locus should be (a).

5. (C)

$$Y_{12} = \left.\frac{I_1}{E_2}\right|_{V_1=0}$$

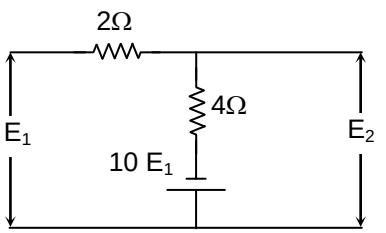
$$E_2 = -20\, I_1,$$

$$Y_{12} = \frac{I_1}{E_2} = -\frac{1}{20} = -0.05\ \text{℧}$$



6. (C)

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0} \qquad Z_{21} = \left.\frac{V_2}{I_1}\right|_{I_2=0}$$



$$Z_{11} = \frac{6}{11}\Omega;\ Z_{21} = -\frac{16}{11}\Omega$$

$$E_2 = -10\, E_1 + 4\, I_1$$

$$E_1 = 6\, I_1 - 10\, E_1$$

$$\frac{E_1}{I_1} = \frac{6}{11},\quad E_2 = \frac{-60 I_1}{11} + 4 I_1 = \frac{-16 I_1}{11}$$

$$\Rightarrow \frac{E_2}{I_1} = \frac{-16}{11}$$

2002

1. (A)

KVL gives

$$20 = 5I + \left(\frac{V_1}{5} + I\right) 5 = 5I + (4 + I)\, 5$$

$\therefore$ I = 0 Amp

Power Delivered = $-\left(\frac{V_1}{5}\right).V_1 = -4 \times 20 = -80$

$\Rightarrow$ Delivers 80 watts

2. (B)

$$R_L = \frac{1}{\frac{1}{20}+\frac{1}{40}} = \frac{40}{3}\,\Omega$$

3. (D)

$$\text{Power} = \frac{3V_L}{\sqrt{3}}\frac{I_L}{\sqrt{3}}\cos\phi$$

$$I_L = \frac{V_L}{\sqrt{3}Z_L}, \quad V_L = 400V$$

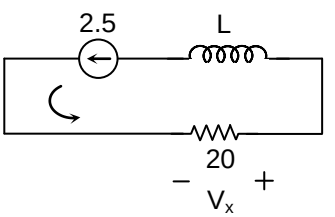
$$1500 = \frac{V_L^2}{Z_L}\cos\phi \quad \Rightarrow \quad Z_L \simeq 90$$

$$\phi = \angle -\cos^{-1}(0.844) = \angle -32.44$$

$$\therefore Z = 90 \angle -32.44^\circ\ \Omega$$

4. (C)

$V_x = 0.25 \times 20 = -50V$



2003

1. (A)

Given n = no. of nodes = 4

b = no. of branches = 7

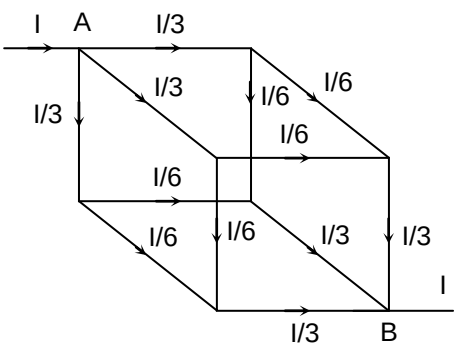
$\therefore$ No. of equations = b – n + 1 = 7 – 4 + 1 = 4

No. of equation = n – 1 = 3 (KCL)

$\therefore$ Minimum No. = 3

2. (A)

The current I will be distributed in the cube branches with the symmetry.



Then, $V_{AB} = \left(\frac{1}{3}+\frac{1}{6}+\frac{1}{3}\right) \times I$

$$R_{AB} = \frac{V_{AB}}{I} = \frac{\frac{5}{6}I}{I} = \frac{5}{6}\Omega$$

3. (C)

By final value theorem,

$$\lim_{\substack{s\to 0\\ t\to\infty}} I(s)\,.\,s \quad \Rightarrow \quad \lim_{s\to 0}\frac{2\times s}{(1+s)} = 2$$

4. (A)

At $t = 0^+$ inductor is open, voltage across capacitor = –V and resistance = 2R

$$i = \frac{-V}{2R}$$

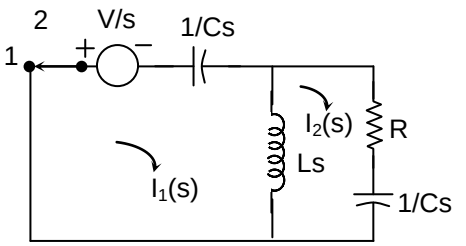
5. (D)

Applying KVL

$$I_1(s)\left\{\frac{1}{Cs}+Ls+R\right\} - I_2Ls = \frac{-V}{s}$$

$$I_2(s)\left\{R+\frac{1}{Cs}+Ls\right\} - I_1Ls = 0$$

$$\begin{bmatrix} R+Ls+\frac{1}{Cs} & -Ls \\ -Ls & R+Ls+\frac{1}{Cs} \end{bmatrix}\begin{bmatrix} I_1(s) \\ I_2(s) \end{bmatrix} = \begin{bmatrix} \frac{-V}{s} \\ 0 \end{bmatrix}$$



6. (B)

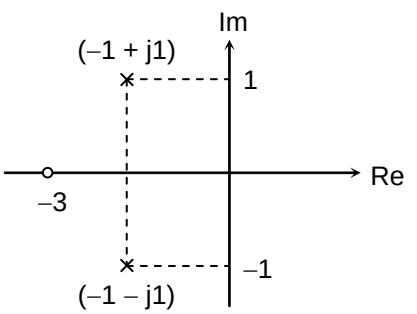
From the figure, $Z(s) = \dfrac{K(s+3)}{(s+1+i)(s+1-i)}$

$$= \frac{K(s+3)}{\left[(s+1)^2+1\right]}$$

As, Z(0) = 3 then, $\dfrac{3K}{2} = 3$

or K = 2

then, $Z(s) = \dfrac{2(s+3)}{(s+1)^2+1} = \dfrac{2(s+3)}{s^2+2s+2}$



7. (B)

$$Q_1 = \frac{\omega L_1}{R_1} = \frac{1}{\sqrt{L_1C_1}}\frac{L_1}{R_1}$$

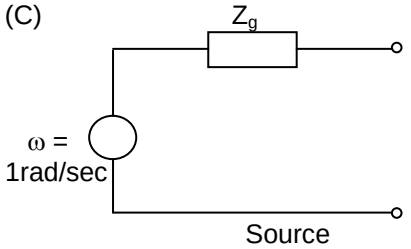
$$Q_2 = \frac{1}{\sqrt{L_2C_2}}\frac{L_2}{R_2}$$

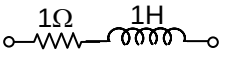
$$C_2 = 2C_1,\quad L_2 = 2L_1\quad R_1 = 2R_2$$

$$\frac{Q_2}{Q_1} = 2$$

$$\frac{Q_1}{2} = Q_2 \quad \Rightarrow 50$$

8. (C)



Source impedance $[Z_g] =$ 

$$[Z_g] = R + jX_L$$
$$= R + j\,\omega L$$
$$= 1 + j\,(1 \times 1) = 1 + j\,\Omega$$

$\therefore$ By conjugate Matching, load that will absorb maximum power is

$$Z_L = 1 - j\,\Omega$$
$$= R - k\,X_C$$
$$= R - j\left(\frac{1}{\omega C}\right)$$

$\therefore$ $\boxed{R = 1}$

$\omega_C = 1$, $C = 1$

i.e. 1 Ω resistance in series with 1 F capacitor.

9. (A)

10. (C)

$i(t) = i_1(t) + i_2(t)$

$$i(t) = \frac{10\sqrt{2}\cos(t + 10°)}{R_1 + j\omega_1 L_1} + \frac{10\sqrt{5}\cos(2t + 10)}{R_1 + j\omega_2 L_1}$$

$$i(t) = \frac{10\sqrt{2}\cos(t + 10°)}{1 + j} + \frac{10\sqrt{5}\cos(2t + 10)}{1 + 2j}$$

$$i(t) = \frac{10\sqrt{2}\cos(t + 10°)}{\sqrt{2}\tan^{-1}(1)} + \frac{10\sqrt{5}}{\sqrt{5}}\frac{\cos(2t + 10^o)}{\tan^{-1}(2)}$$

$$= \frac{10\sqrt{2}\cos(t + 10^o)}{\sqrt{2}\tan^{-1}(1)} + \frac{10\sqrt{5}\cos(2t + 10^o)}{\sqrt{5}\tan^{-1}(2)}$$

$$= 10\cos(t - 35^o) + 10\cos(2t + 10^o - \tan^{-1} 2)$$

11. (C)

Applying KVL $\quad 2i(t) + 2\dfrac{di(t)}{dt} + \dfrac{1}{1}\displaystyle\int i\,dt = \sin t$

Differentiating $\quad 2\dfrac{d^2i}{dt^2} + 2\dfrac{di}{dt} + i(t) = \cos t$

12. (A)

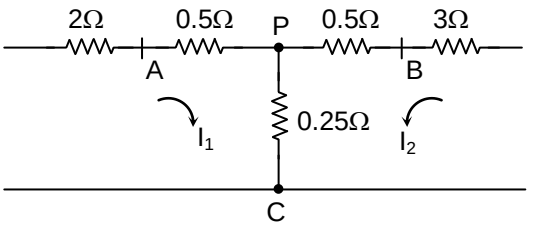
$$Z_{AP} = \frac{Z_{AB}\ Z_{AC}}{Z_{AB} + Z_{AC} + Z_{BC}} = \frac{2}{4} = 0.5\Omega$$

$$Z_{PB} = 0.5\Omega$$

$$Z_{PC} = 0.25\Omega$$

$$V_1 = 2.75 I_1 + 0.25\, I_2$$

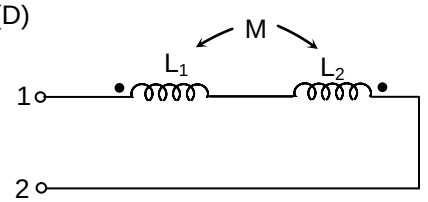
$$Z_{11} = 2.75\Omega, \quad Z_{22} = 0.25\Omega$$

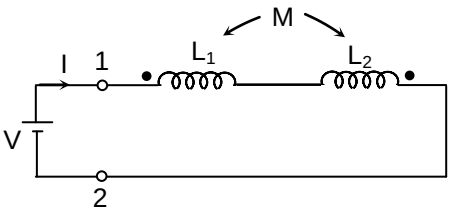


2004

1. (B)

2. (D)

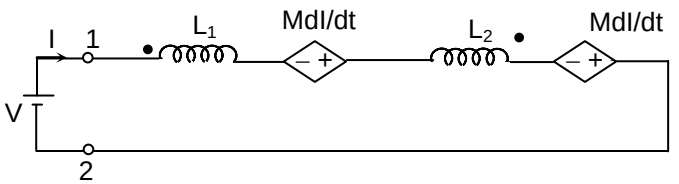




Equivalent inductance (L_{eq})

$$V = L_{eq}\frac{dI}{dt}$$

Now using concept of dot, we assign voltage polarity to both L_1 and L_2



now applying KVL from 2 to 1, we get

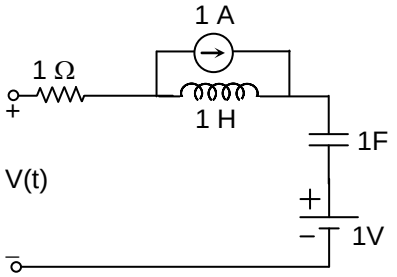
$$V = -M\frac{dI}{dt} + L_2\frac{dI}{dt} - M\frac{dI}{dt} + L_1\frac{dI}{dt}$$

$$V = \left[L_1 + L_2 - 2M\right]\frac{dI}{dt}$$

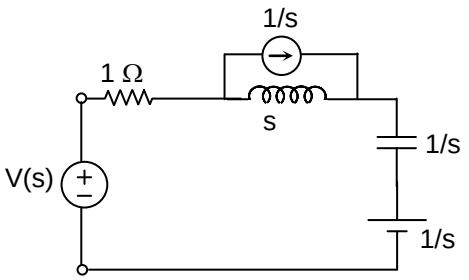
$$\therefore \quad L_{eq} = L_1 + L_2 - 2M$$

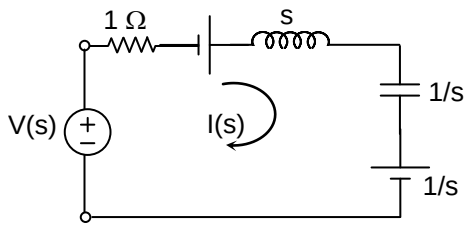
3. (A)

At t > 0,



Taking Laplace transform of the circuit, we get





$$V(s) - I(s) + 1 - SI(s) - \frac{1}{s} - \frac{1}{s}I(s) = 0$$

$$V(s) + 1 - \frac{1}{s} = \left[\frac{1}{s} + s + 1\right]I(s) \qquad \ldots \begin{bmatrix} V(t) = u(t), \\ V(s) = 1/s \end{bmatrix}$$

$$\frac{1}{s} + 1 - \frac{1}{s} = \left[\frac{1}{s} + s + 1\right]I(s)$$

$$1 = \frac{\left[s^2 + s + 1\right]}{s}I(s)$$

$$I(s) = \frac{s}{s^2 + s + 1}$$

4. (D)

Applying KVL,

$$\left(10 \times 10^3\right) v_i(s) + \left(10 \times 10^{-3}\right) s\, v_i(s) + \frac{1}{s\left(100 \times 10^{-6}\right)} V_i(s) = V_c(s)$$

$$\frac{V_o(s)}{V_i(s)} = \frac{\dfrac{1}{s(100\times10^{-6})}}{(10\times10^{3}) + s(10\times10^{-3}) + \dfrac{1}{s(100\times10^{-6})}} = \frac{10^6}{s^2 + 10^6 s + 10^6}.$$

5. (B)

$$H(s) = \frac{10^6}{s^2 + 20s + 10^6}$$

$$= \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + (\omega_n)^2}$$

$$\omega_n^2 = \alpha\ 10^6$$

$$\therefore \quad \omega_n = 10^3 \text{ rad / sec.}$$

$$2\zeta\omega_n = 20$$

$$\zeta = \frac{20}{2\times10^3} = 10\times10^{-3} = 10^{-2} = 0.01$$

$$Q = \frac{1}{2\zeta} = \frac{1}{2\times0.01} = \frac{1}{0.02} = 50$$

6. (A)

$$\frac{d^2y}{dt^2} + 3\frac{dy}{dt} + 2y = 2u(t)$$

Taking Lapalce transform,

$$\left[s^2\,Y(s) - s\,y(0) - y(0)\right] + 3\left[s\,Y(s) - y(0)\right] + 2\ Y(s) = \frac{2}{s}$$

$$\therefore \quad s^2\,Y(s) + 3s\ Y(s) + 2\ Y(s) = \frac{2}{s}$$

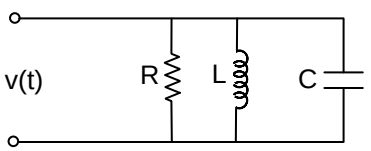
$$Y(s)\left(s^2 + 3s + 2\right) = \frac{2}{s}$$

$$\therefore \quad y(s) = \frac{2}{s(s^2 + 3s + 2)} = \frac{2}{s(s+1)(s+2)}$$

$$= \frac{K_o}{s} + \frac{K_1}{s+1} + \frac{K_2}{s+2} = \frac{1}{s} - \frac{2}{s+1} + \frac{1}{s+2}$$

$$\therefore \alpha\, L^{-1}[ycs] = \left[1 - 2e^{-t} + e^{-2t}\right]u(t)$$

7. (A)



v(t) = sin 2t = 1 $\angle$ 0°

ω = 2 rad/sec

v(t) = i(t) z

$$\therefore \quad z = \frac{1}{y}$$

$$\therefore \quad y = \frac{1}{R} + \frac{1}{jX_L} + \frac{1}{-jX_C}$$

$$\therefore \quad y = \frac{1}{R} - j\frac{1}{X_L} + j\frac{1}{X_C}$$

$$= \frac{1}{(1/3)} - j\frac{1}{\left[2 \times \frac{1}{4}\right]} + j(2 \times 3)$$

y = 3 – j2 + j6 = 3 + j4

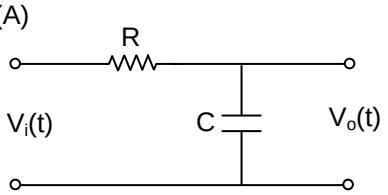
$\therefore$ z = 0.12 – j 0.16 = 0.2 $\angle$ –53.13°

$$\therefore \quad i(t) = \frac{v(t)}{z} = \frac{1\angle 0°}{0.2\angle -53.13°} = 3 + j4 = 5\angle 53.13°$$

$\therefore$ i(t) = 5 $\angle$ 53.13°

i(t) = 5 sin(2t + 53.13°) A

8. (A)



RC = 1 ms

$V_i(t) = \sqrt{2}\sin 10^3 t = \sqrt{2}\angle 0°$

$\omega = 10^3$ rad/sec

$$V_o(t) = \frac{-jX_C V_i(t)}{(R - jX_C)}$$

$$V_o(t) = \frac{\frac{-j}{\omega C} V_i(t)}{\left(R - \frac{j}{\omega C}\right)}$$

$$\therefore \quad V_o(t) = \frac{-j\,V_i(t)}{[\omega(RC) - j]}$$

$$= \frac{-j}{\left[10^3\left[10^{-3}\right] - j\right]}\left(\sqrt{2}\angle 0°\right)$$

$$V_o(t) = 1\angle -45°$$

$$V_o(t) = \sin\left[10^3 t - 45°\right] \text{ V}$$

9. (C)

$$i(t) = \frac{V}{R}\left[1-e^{-t/\tau}\right]$$

$$i(t) = \frac{1}{2}\left[1-e^{-t/0.5}\right]$$

10. (D)

$$Z_{11} = Z_{22} = \frac{Z_a + Z_b}{2}$$

$$= \frac{j2+2}{2} = j+1$$

$$Z_{21} = Z_{12} = \frac{Z_b - Z_a}{2}$$

$$= \frac{2-j2}{2} = 1-j.$$

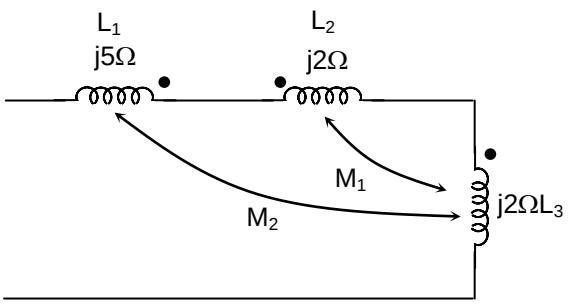
2005

1. (B)

$$\therefore \quad L_{eq} = L_1 + L_2 + L_3 + 2M_1 - 2M_2$$

$$= (5+2+2+20-20)j$$

$$L_{eq} = 9j.$$



2. (C)

$$V_{R_2} = \frac{R_2}{R_1+R_2} = \frac{R}{2R} \times V = 5V$$

$$V_{R_3} = \frac{1.1R}{R+1.1R} \times V$$

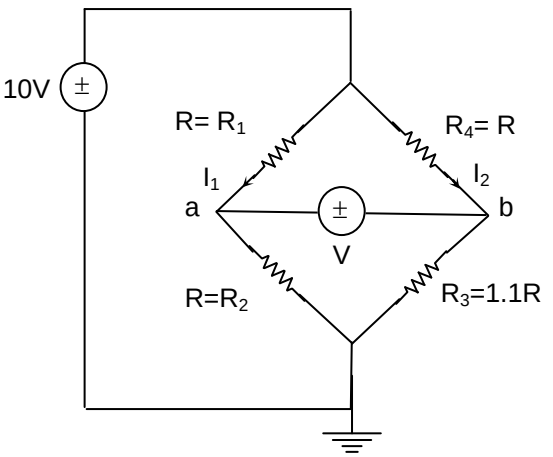
$$= \frac{1.1R}{2.1R} \times V$$

$$= 5.238V$$

$$V = V_{R_2} - V_{R_3}$$

$$= -0.238V$$

$\therefore$ Option (C) is correct answer.

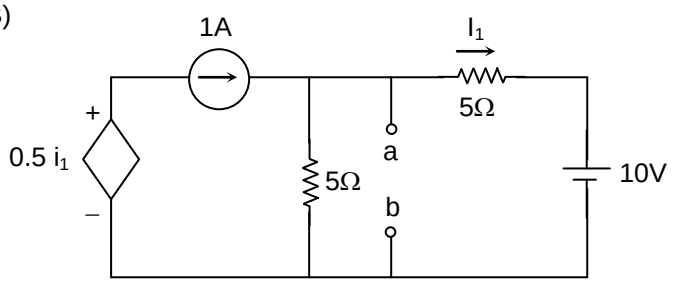


3. (C)

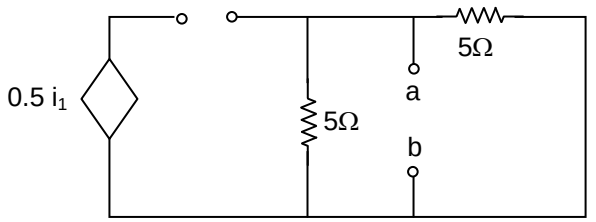
Maximum power that can be transferred is given by maximum power transfer theorem.

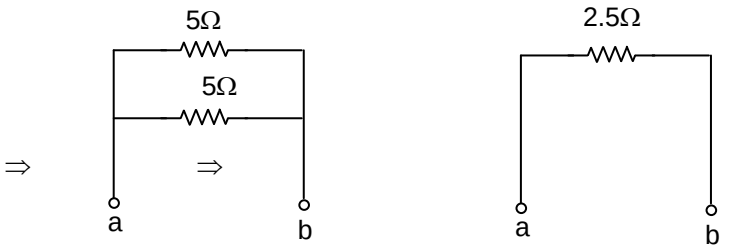
$$\text{Power} = \frac{V^2}{4R} = \frac{(10)^2}{4(100)} = \frac{100}{400} = 0.25\,W$$

4. (B)

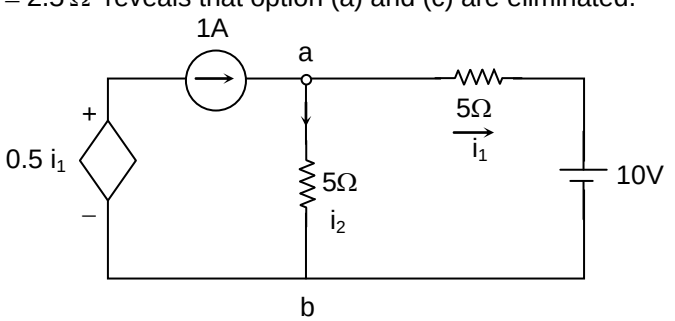


To find thevenin equivalent resistance.





$R_{TH} = 2.5\,\Omega$ reveals that option (a) and (c) are eliminated.



Consider 1A alone, $\therefore$ current will split equally in two branches of 5Ω alone

$\therefore$ $i_1 = i_2 = 0.5A$

$\therefore$ $5 \times 0.5 = 2.5$ V

Now $V_{TH} = 10 - 2.5 = 7.5$ V

$\therefore$ Option (B) is correct option.

5. (A)

Input voltage $V_i = 3$ V

Capacitor will get charged to 3 V

$V_C(t) = V_1\left[1 - e^{-t/z}\right]$ $\qquad z = RC = 1 \times 10^3 \times 0.1 \times 10^{-6} = 10^{-4}$

$= V_1\left[1 - e^{-t/10^{-4}}\right]$

$V_C(t) = 3\left[1 - e^{-t/10^{-4}}\right]$

At t = 2 sec,

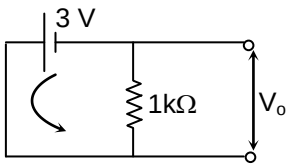
$V_C(2) = 3\left[1 - e^{-2/10^{-4}}\right]$

$\boxed{V_C(2) = 3\text{ V}}$

At t = 2 sec, input voltage changes to zero volts.
But voltage across capacitor cannot change instantaneously

$\therefore \quad \boxed{V_C(2^+) = 3\text{ V}}$

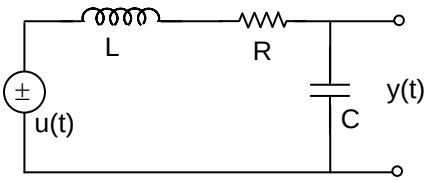
At t = 2 sec,



$\boxed{V_o = -3\text{V}}$

6. (C)

The RLC circuit is as shown below,



Damping ratio $\xi = \frac{R}{2}\sqrt{\frac{C}{L}}$

For no oscillations to occur, $\varepsilon > 1$ is the ideal condition.

$$\therefore \quad \frac{R}{2}\sqrt{\frac{C}{L}} > 1$$

$$\therefore \quad R\sqrt{\frac{C}{L}} > 2$$

$$R > 2\sqrt{\frac{L}{C}}$$

Hence correct option is (C)

7. (B)

In series RLC circuit, the resonant frequency is given by.

$$\omega_n = \frac{1}{\sqrt{LC}}$$

$$= \frac{1}{\sqrt{1 \times \frac{1}{400} \times 10^{-6}}}$$

$$\omega_n^2 = 400 \times 10^6$$

$$\omega_n = 2 \times 10^4$$

$$f_n = \frac{1}{2\pi} \times 2 \times 10^4 = \frac{1}{\pi} \times 10^4 .$$

8. (A)

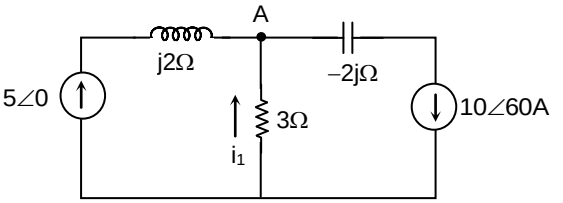
Applying KCL at node A we get,

$$5\angle 0 + i = 10\angle 60$$

$\therefore \quad 5 + 0j + i = 5 + j8.66$

$\therefore \quad i = +j8.66$

$\therefore \quad i = \frac{10\sqrt{3}}{2} \angle +90^\circ$ Amps.



9. (D)

Hybrid parameters:

$$V_1 = h_{11} I_1 + h_{12} V_2$$

$$I_2 = h_{21} I_1 + h_{22} V_2$$

$$h_{11} = \left.\frac{V_1}{I_1}\right|_{V_2=0} \qquad h_{12} = \left.\frac{V_1}{V_2}\right|_{I_1=0}$$

$$h_{21} = \left.\frac{I_2}{I_1}\right|_{V_2=0} \qquad h_{22} = \left.\frac{I_2}{V_2}\right|_{I_1=0}$$

$$V_1 = 10\, I_1 + V_2$$

$$I_2 + I_1 = \frac{V_2}{20}$$

or $\quad I_2 = -I_1 + \frac{V_2}{20}$

$\therefore \quad h_{11} = 10$

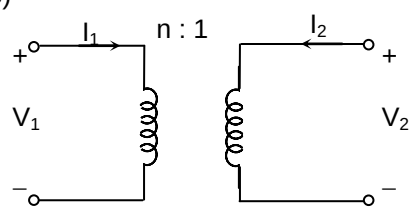
$h_{12} = 1$

$h_{21} = 1$

$h_{22} = \frac{1}{20} = 0.05$

$\therefore$ Option (D) is correct

10. (B)



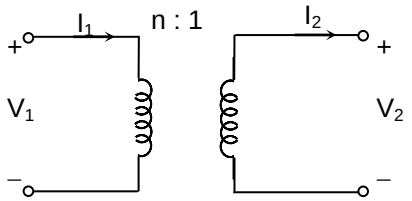
ABCD parameters

$V_1, I_1 = f(V_2, -I_2)$

$V_1 = A\,V_2 - B\,I_2$

$I_1 = C\,V_2 - D\,I_2$

For transformer,



We know that,

$$\frac{V_1}{V_2} = \frac{n}{1} = \frac{I_2}{I_1}$$

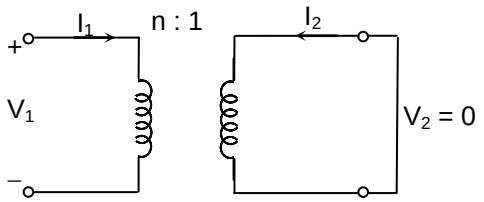
But for the given transformer in question, I_2 is in opposite direction

$$\therefore \quad \frac{V_1}{V_2} = \frac{n}{1} = \frac{-I_2}{I_1}$$

$I_1 = CV_2 - DI_2$

$$D = \left.\frac{I_1}{-I_2}\right|_{V_2=0}$$

$\therefore$ Short–circuiting port (2), we have



$$\frac{V_1}{V_2} = \frac{n}{1} = \frac{-I_2}{I_1}$$

$$\therefore \quad \frac{n}{1} = \frac{-I_2}{I_1}$$

$$\therefore \quad D = \frac{-I_1}{I_2} = \frac{1}{n} \quad \therefore \quad x = \frac{1}{n}$$

11. (D)

The general expression for RL driving point impedance function is

$$Z = Z_1 + \cfrac{1}{y_2 + \cfrac{1}{y_4 + \cfrac{1}{Z_5 + \cfrac{1}{y_6}}}}$$

In RC impedance fun c, nearest critical frequency to the origin is a pole and a farthest critical frequency to the origin is zero.

2006

1. (A)

Initial current = $i(0^-)$

$$L \times i(0^-) = 1\text{mV}$$

$$\Rightarrow \quad 2 \times 10^{-3} \times i(0^-) = 1 \times 10^{-3}$$

$$\Rightarrow \quad i(0^-) = \frac{1}{2} = 0.5\text{ A}$$

2. (B)

For no phase – distortion, the phase spectrum $\phi(\omega)$ should be linear with the frequency

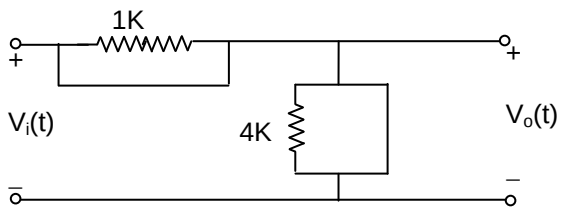
3. (B)

$V_i(t) = 10\ u(t)$

All the capacitors are initially uncharged

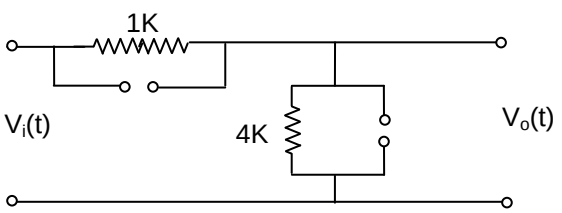
At $t = 0^+$

Capacitors will be short circuited



$\therefore$ Output is short circuited, $V_o(0^+) = 0$

At $t = \infty$, capacitors will be open circuited.



$$V_o(t) = \frac{V_i(t) \times 4K}{[4+1]K}$$

$$V_o(t) = \frac{4}{5} V_i(t) = \frac{4}{5} 10\ u(t)$$

At $t = \infty$, $V_o(\infty) = \frac{4}{5} \times 10 = 8\text{ V}$

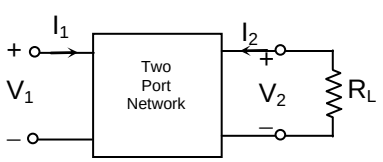
$\therefore$ Check from the options,

for $t = 0^+$ & $t = \infty$

$V_o = 0$ $\quad$ $V_o = 8$

Only option (B) satisfies both the conditions.

4. (D)



$$\begin{bmatrix} V_1 \\ I_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} V_2 \\ -I_2 \end{bmatrix}$$

$\therefore \quad V_1 = AV_2 - BI_2 \qquad \ldots(1)$

$\quad I_1 = C\,V_2 - DI_2 \qquad \ldots(2)$

Now, port 2 is terminated by R_L, $V_2 = -I_2R_L$

Input impedance seen at port 1 is given by

$$\frac{V_1}{I_1} = z_1$$

$\therefore \quad V_2 = -I_2\,R_L$

Substituting $V_2 = -\,I_2R_L$ in equation (1) and (2)

$V_1 = A[-I_2R_L] - BI_2$

$V_1 = -[AR_L + B]\,I_2 \qquad \ldots(3)$

$I_1 = C[-I_2R_L] - DI_2$

$I_1 = -[CR_L + D]\,I_2$

$$\therefore \quad I_2 = -\frac{I_1}{[CR_L + D]}$$

Substituting $I_2 = -\dfrac{I_1}{[CR_L + D]}$ in equation (3) we get

$$V_1 = -\left[AR_L + B\right]\frac{(-I_1)}{[CR_L + D]}$$

$$\therefore \quad \frac{V_1}{I_1} = z_1 = \left[\frac{AR_L + B}{CR_L + D}\right]$$

5. (B)

$V_1 \;= I_1r_e$

$V_2 \;= \beta I_2r_0 + r_0\,I_2$

Comparing with

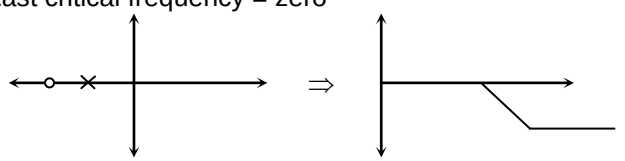
$V_1 \;= Z_{11}\,I_1 + Z_{12}\,I_2$

$V_2 \;= Z_{21}\,I_1 + Z_{22}\,I_2$

6. (B)

First critical frequency = pole

Last critical frequency = zero



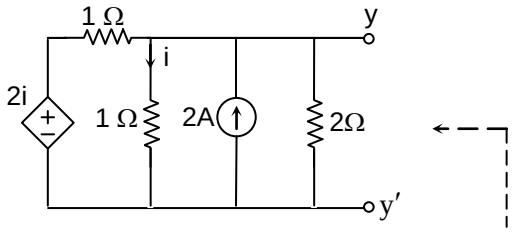
Bode plot

This plot corresponds to a low–pass filter which will be satisfied by RC network only.

7. (A)

2007

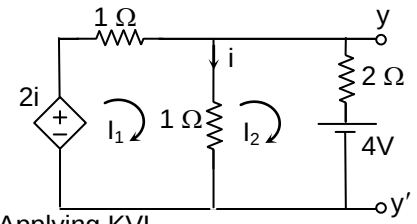
1. (D)



Since, the given circuit contains dependent voltage source we cannot short circuit the voltage source

To find V_{TH}

Converting the current source into voltage source



Applying KVL,

$2i - I_1 - I_1 + I_2 = 0$

$2I_1 - I_2 = 2i$ …(1)

$-2I_2 - 4 - I_2 + I_1 = 0$

$I_1 - 3I_2 = 4$ …(2)

Also, $I_1 - I_2 = i$

$\therefore$ $2I_1 - I_2 = 2i$

$2I_1 - I_2 = 2(I_1 - I_2)$

$2I_1 - I_2 = 2I_1 - 2I_2$

$I_2 = 0$

$\therefore$ $V_{TH} = 4 + 2I_2$

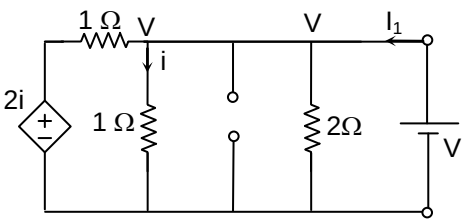
$= 4 + 2(0)$

$\boxed{V_{TH} = 4\text{ V}}$

To find R_{TH}

We cannot short circuit the dependent voltage source to find R_{TH}. But we can open circuit the independent current source of 2A.

We apply a voltage source V between y and y′ and measure current through it.



$R_{TH} = \dfrac{V}{I_1}$

Apply KCL,

$$\frac{V-2i}{1}+\frac{V}{1}+\frac{V}{2}=I_1$$

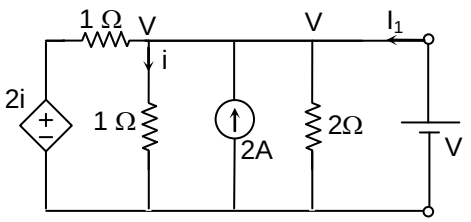
Also, $\dfrac{V}{1}=i \qquad \therefore \quad V = i$

$$-V + V + \frac{V}{2}=I_1$$

$$\therefore \quad \frac{V}{I_1}=2=R_{TH}$$

$\therefore \quad \boxed{R_{th} = 2\ \Omega}$

Alternative :



Apply KCL,

$$\frac{V-2i}{1}+\frac{V}{1}+\frac{V}{2}=I_1+2$$

$V = I, \quad -V + V + \dfrac{V}{2} = I_1 + 2$

$\therefore \qquad \boxed{V = 2I_1 + 4}$

To find V_{th},

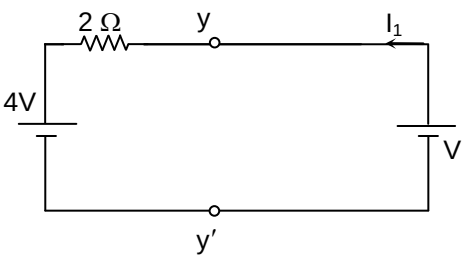
Open circuit $y = y'$

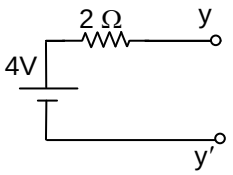
$\therefore \qquad I_1 = 0$

$V = 2I_1 + 4$

$V_{th} = V\big|_{I_1=0} = 2(0) + 4 = 4V$

Equation [$V = 2I_1 + 4$] can be expressed in diagrammatic form as

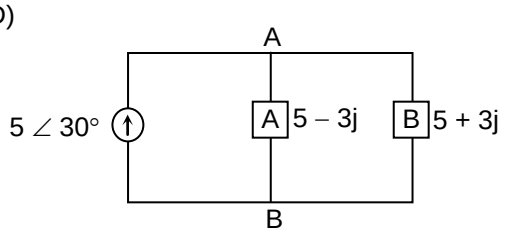




$\therefore \quad V_{th} = 4\ V$
$\quad\quad R_{th} = 2\ \Omega$

2. (D)
For maximum power transfer at load $Z_L = Z_S$ is the required condition
$Z_L = R_S - jX_S$

3. (D)



Load A and load B in parallel
$\therefore$ Their equivalent resistance is

$$\left[\left(\frac{1}{5-3j}\right)+\left(\frac{1}{5+3j}\right)\right]^{-1} = 3.4$$

$\therefore \quad V_{AB} = (5 \angle 30°) \times (3.4)$
$\quad\quad = 17 \angle 30°$

4. (A)
$i_C(t) = i_{C\ final} + (i_{C\ initial} - i_{C\ final})\, e^{-t/RC}$
$R' = 201120 = 10\ K\Omega$
$R'C = 10\ K \times 4\mu = 40$ m.
$i_{C\ initial} = 0$
$i_{C\ final} = 10/20\ k = 0.5$ mA
$i_C\ (t) = 0.5\ e^{-t/40m}$ mA
$= 0.5\ e^{-25t}$ mA

2008

1. (B)

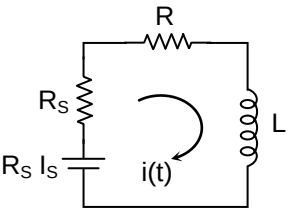
2. (A)
For finding Thevenin equivalent
Short voltage source
Open circuit source

Now Thevenin equivalent = $\left(1+\frac{1}{S}\right) \| (1+S) = \dfrac{1+\frac{1}{S}+S+1}{1+\frac{1}{S}+1+S} = 1$

3. (B)

Circuit can be transformed as shown, after the switch is closed.

$\therefore$ by KVL



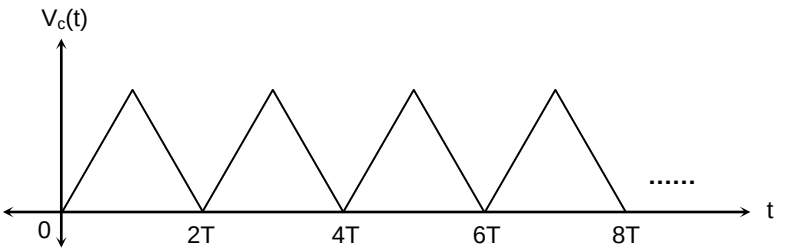
$$R_S I_S - R_S\, i(t) - L\frac{di(t)}{dt} = 0$$

$\therefore$ at $t = 0^+$; and knowing $i(0^+) = 0$; we get

$$\frac{di}{dt}(0^+) = \frac{R_S I_S}{L} \text{ A / S}$$

4. (C)

The capacitor is alternately charging and discharging also resistance 0.5 Ω and 1Ω will not affect the charging and discharging $\because$ they are in series with constant current source, also capacitor charges and discharges linearly, whose waveform is shown below.



$\therefore$ The signal which gives this waveform is C.

5. (D)

$$V_C(s) = \frac{1/s}{1+s+\frac{1}{s}} = \frac{1}{s^2+s+1} = \frac{1}{\left(s^2+s+\frac{1}{4}\right)+\frac{3}{4}}$$

$$= \frac{1}{\left(s+\frac{1}{2}\right)^2+\left(\frac{\sqrt{3}}{2}\right)^2} = \frac{2}{\sqrt{3}}\cdot\frac{(\sqrt{3}/2)}{(s+0.5)^2+\left(\frac{\sqrt{3}}{2}\right)^2}$$

$$\Rightarrow\ V_C(t) = \frac{2}{\sqrt{3}}e^{-0.5t}\sin\left(\frac{\sqrt{3}}{2}t\right)$$

6. (B)

$$i(t) = \frac{CdV_C t}{dt} = \frac{2}{\sqrt{3}}\frac{d}{dt}\left[e^{-0.5t}\sin\frac{\sqrt{3}}{2}t\right]$$

$$= \frac{2}{\sqrt{3}}\left[-0.5e^{-0.5}\sin\frac{\sqrt{3}}{2}t+\frac{\sqrt{3}}{2}e^{-0.5t}\cos\frac{\sqrt{3}}{2}t\right]$$

$$= e^{-0.5\,t}\left[\cos\frac{\sqrt{3}}{2}t-\frac{1}{\sqrt{3}}\sin\frac{\sqrt{3}}{2}t\right]$$

$$\therefore\ V_R(t) = i(t)$$

7. (D)

$$\frac{1}{y(s)} = \frac{1}{SL} + SC + \frac{1}{R} = \frac{R + S^2RLC + SL}{SRL}$$

$$\therefore \quad Z(S) = \frac{SRL}{S^2RLC + SL + R} = \frac{S/C}{S^2 + \frac{1}{RC}S + \frac{1}{LC}}$$

comparing with this, we get

$$\frac{1}{C} = 0.2 \quad \Rightarrow \quad C = 5F$$

$$\frac{1}{RC} = 0.1 \quad \Rightarrow \quad RC = 10 \quad \therefore \quad R = 2\Omega$$

$$\frac{1}{LC} = 2 \quad \Rightarrow \quad L = 0.1\ H$$

8. (C)

$V_1 = Z_{11}I_1 + Z_{12}I_2$

$V_2 = Z_{21}I_1 + Z_{22}I_2$

Using the given information : S_1-open, S_2-closed

$4.5 = Z_{12}1 \Rightarrow Z_{12} = 4.5$

$1.5 = Z_{22}1 \Rightarrow Z_{22} = 1.5$

S_1-closed, S_2-open

$6 = Z_{11}1 \Rightarrow Z_{11} = 1.5$

$6 = Z_{21}4 \Rightarrow Z_{21} = 1.5$

Thus Z matrix = $\begin{bmatrix} 1.5 & 4.5 \\ 1.5 & 1.5 \end{bmatrix}$

9. (A)

$$V_1 = Z_{11}I_1 + Z_{12}I_2 \qquad \ldots(1)$$

$$I_2 = \frac{-Z_{21}I_1 + V_2}{Z_{22}}$$

$$I_2 = \frac{-Z_{21}}{Z_{22}}I_1 + \frac{1}{Z_{22}}V_2 \qquad \ldots(2)$$

Substitute (2) in (1) to get V_1 in terms of I_1 and V_2

Thus H matrix = $\begin{bmatrix} -3 & 3 \\ -1 & 0.67 \end{bmatrix}$

2009

1. (A)

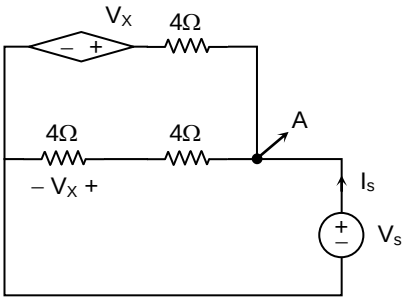
$\because$ 60 V source is absorbing power, the direction of current I_V flowing through it is $\downarrow$ (i.e. from the positive terminal to negative terminal)

$\therefore \quad I + I_V = 12 \Rightarrow \quad I < 12$ A

2. (C)

We will find R_{th} and then for maximum power transfer $R_L = R_{th}$, to find R_{th}, we make all independent sources zero and V_S in place of R_L then $R_{th} = \frac{V_s}{I_s}$.

Now KCL at node A, we get



$$\frac{V_s}{8} + \frac{V_s - V_x}{4} = I_S \qquad \ldots(1)$$

also $\quad V_x = \frac{V_s}{8}.4 = \frac{V_s}{2}$

$\therefore$ Putting in (1), we get

$$V_s + 2V_s - V_s = 8\,I_s$$

$$\therefore \qquad 2V_s = 8I_s \Rightarrow \frac{V_s}{I_s} = R_{th} = \frac{8}{2} = 4\Omega$$

3. (C)

$$z_C = \frac{R_L \frac{1}{SC}}{R_L + \frac{1}{SC}}$$

$$z_C = \frac{R_L}{SCR_L + 1}$$

$$\frac{V_o(S)}{V_i(S)} = \frac{\frac{R_L}{SCR_L + 1}}{\frac{R_L}{SCR_L} + R}$$

$$= \frac{R_L}{R_L + RSCR_L + R} = \frac{R_L}{SCRR_L + R + R_L} \Rightarrow R_L = R$$

4. (A)

$$I = \frac{20}{z_{total}} = \frac{20}{\sqrt{8^2 + 6^2}} = 2\text{Amp} \qquad [\because\ Z_{total} = Z_s + Z_c = 8 + 6j]$$

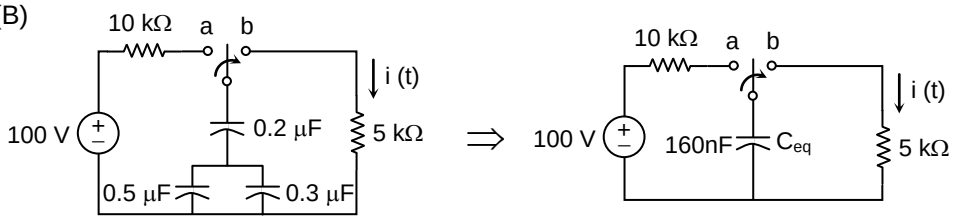
$\therefore$ Reactive power = Power reactive element of load = $4 \times 2 = 8$ VAR

5. (A)

… ($\because$ sinusoidal term will become zero at $t \to \infty$ and steady state does not depend on initial conditions)

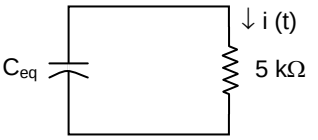
$$\therefore \quad i(t)\big|_{t\to\infty} \longrightarrow \frac{V_0}{R}$$

6. (B)



$C_{eq} = (0.5\ \mu + 0.3\ \mu) \parallel 0.2\ \mu$
$= 160$ nF

and $V_{Ceq}(0^-) = V_{Ceq}(0^+) = 100$ V



$\therefore$ Circuit at t > 0 becomes

$\because$ this is a simple RC circuit

$i(t) = I(0)\, e^{-t/T}$ … [$I_0 = 20$ mA, T = RC = 800 μ sec $\Rightarrow \frac{1}{T} = 1250 s^{-1}$]

$\therefore$ $i(t) = 20\, e^{-1250t} u(t)$ mA

2010

1. (A)

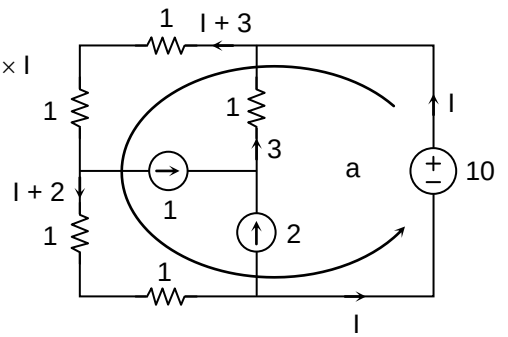
Power supplied by voltage source = V × I

Writing KVL in loop

$10 = 2 \times (I + 3) + 2 \times I + 2$

$\Rightarrow I = 0$

So P = 0 W



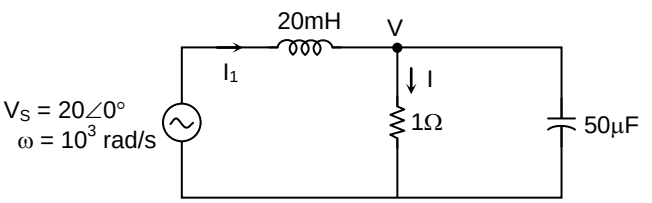
2. (A)

Nodal at V,

$$\frac{V-20}{LS} + \frac{V}{R} + \frac{V}{1/CS} = 0$$

$$V\left[\frac{1}{LS} + \frac{1}{R} + CS\right] = \frac{20}{LS}$$

$$V = \frac{\frac{20}{LS}}{\frac{1}{LS} + \frac{1}{R} + CS}$$



So $I = \frac{V}{R} = \frac{V}{I} = V$

Putting S = jω and L, R, C

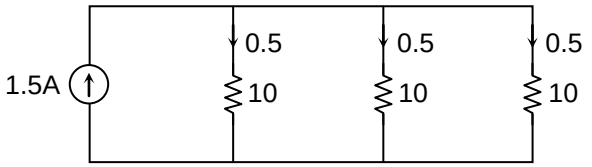
$V = -j = I$

3. (D)

For a parallel RLC circuit, at resonance, $\frac{1}{Z} = \frac{1}{R} + \frac{1}{X_L} + \frac{1}{X_C}$ which is maximum at resonance.

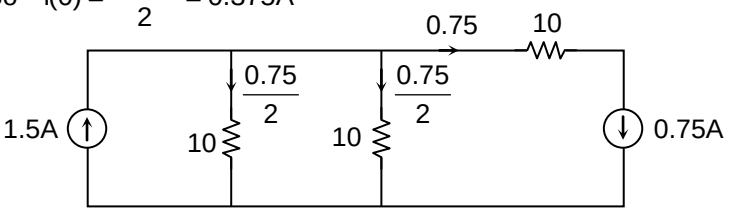
4. (A)

i(∞) = 0.5A as L will be come shorted At t = ∞



Also at t = 0, L will be current source of 0.75 A

So $i(0) = \frac{-75A}{2} = 0.375A$



So knowing i(0) and i(∞) option A is answer.

5. (A)

Short circuit admittance for π network can directly be written as :

$$Y = \begin{vmatrix} \frac{1}{0.5}+\frac{1}{0.5} & -\frac{1}{0.5} \\ -\frac{1}{0.5} & \frac{1}{0.5}+\frac{1}{0.5} \end{vmatrix} = \begin{bmatrix} 4 & -2 \\ -2 & 4 \end{bmatrix}$$

2011

1. (A)

$$\underline{I}_N = \underline{I}_{shPQ} = 16\angle 0^\circ \times \frac{25}{(40+j30)} = 8\angle -\tan^{-1}\left(\frac{3}{4}\right) = (6.4 - j4.8)A$$

2. (C)

$R_L = R_{th} = 15\ \Omega$

3. (B)

$\underline{Z}_S = 7\angle 0^\circ\ \Omega$ (Using Y – Δ transformation)

$$\underline{I} = \frac{14\angle 0^\circ}{7\angle 0^\circ} = 2\angle 0^\circ\ A$$

4. (A)

$$\omega = \frac{1}{RC}$$

$$\frac{V_0}{V_i} = \frac{z_P}{z_P + z_S} \quad \ldots\ldots z_P = \frac{R.\frac{1}{SC}}{R+\frac{1}{SC}} = \frac{R}{1+SRC}, \quad z_S = R + \frac{1}{SC}$$

For $S = j\omega = \frac{j}{RC}$, we get

$$z_P = \frac{R}{1+j}, \quad z_S = R(1-j)$$

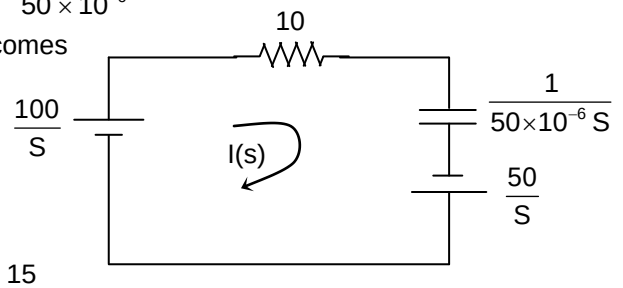
$$\therefore \frac{V_0}{V_i} = \frac{\frac{R}{1+j}}{\frac{R}{1+j} + R(1-j)} = \frac{1}{1+(1-j^2)} = \frac{1}{3}$$

$$V_0 = \frac{1}{3}V_i = \frac{V_P}{3}\cos\left(\frac{t}{RC}\right)$$

5. (A)

$$V_c(0^+) = \frac{I_c}{C} = \frac{2.5\times10^{-3}}{50\times10^{-6}} = 50$$

$\therefore$ Circuit becomes



$$\Rightarrow I(s) = \frac{15}{S + 2\times10^3}$$

$\Rightarrow i(t) = 15 \exp(-2\times10^3 t)$

6. (D)

$N_1 = 100V + 25I_1$; $V_2 = -100I_2$

$I_2 = Y_3V_1 + Y_4V_2 \Rightarrow -0.01V_2 = 0.01V_1 + 0.1V_2 \Rightarrow \frac{V_2}{V_1} = \frac{-1}{11}$

2012

1. (C)

Required I_2

Now,

$I_2 - I_1 = 1$ ….(1)

$I_3 - I_4 = 1$ ….(2)

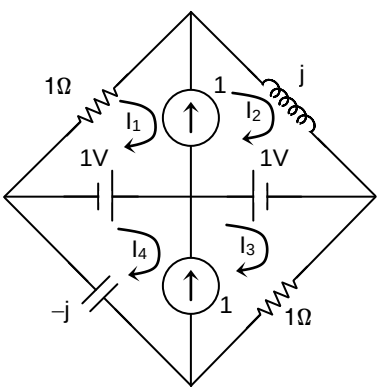
Also by KVL in suppermesh

$-1 - I_1 - jI_2 + 1 = 0$

$I_1 + jI_2 = 0$ ….(3)

Solving (1) & (3) for I_2, we get

$$I_2 = \frac{1}{1+j}\text{Amp}$$

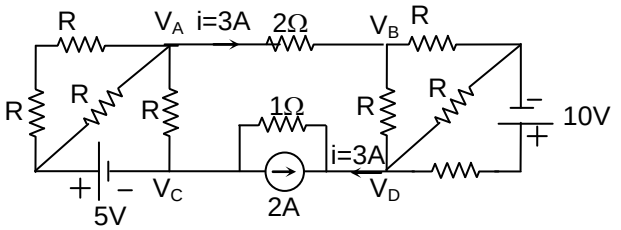


2. (A)

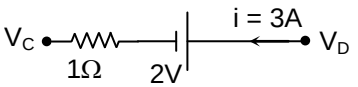
$V_A - V_B = 6$ V

$\therefore$ Current through 2 Ω resistor is,

$$i = \frac{V_A - V_B}{2} = \frac{6}{2} = 3A$$



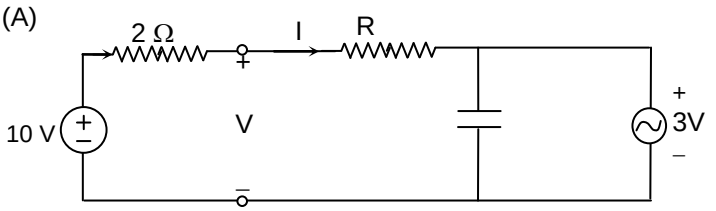
Since current I = 3A is entering the network on the right, the same current i = 3A will leave the network.
And converting the current source into voltage source.



$\therefore \quad V_C - V_D = -2 - 3\times1$

$V_C - V_D = -5V$

3. (A)



Power delivered to circuit B is,

$P = VI$

By applying KVL to circuit B, we get

$V = 3 + IR$

$\therefore \quad P = (3 + IR)\, I$

Now Applying KVL to the whole circuit, we get

$10 - 2I - RI - 3 = 0$

$$I = \frac{7}{(R+2)}$$

$$\therefore \quad P = \left[3 + \frac{7R}{R+2}\right]\frac{7}{(R+2)}$$

$$P = \frac{[10R+6]7}{(R+2)^2} = \frac{70R+42}{(R+2)^2}$$

For maximum power transfer to circuit B,

$$\frac{dP}{dR} = 0$$

$$\therefore \quad \frac{dP}{dR} = \frac{(R+2)^2[70] - (70R+42)[2(R+2)]}{(R+2)^4} = 0$$

$\therefore \quad (R + 2)^2\, 70 = 2[R + 2]\,[70R + 42]$

$70R + 140 = 140\,R + 84$

$56 = 70\,R$

$\therefore \quad R = 0.8\ \Omega$

4. (B)

$$P_{arg} = \frac{1}{2}R_e\left[VI^*\right]$$

$$= \frac{1}{2}Re\left[IZI^*\right]$$

$$= \frac{1}{2}I^2\,Re[z]$$

$$= \frac{1}{2} \times 25 \times 4$$

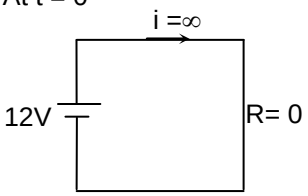
$$= 50\text{ W}$$

5. (D)

$V_{C1}(0^-) = 12$ V $\qquad$ $V_{C2}(0^-) = 0$ V

$\therefore$ $V_{C1}(0^+) = 12$ V $\qquad$ $\therefore$ $V_{C2}(0^+) = 0$ V

At $t = 0^+$



$$V = i\,R$$

$$i = \frac{V}{R}$$

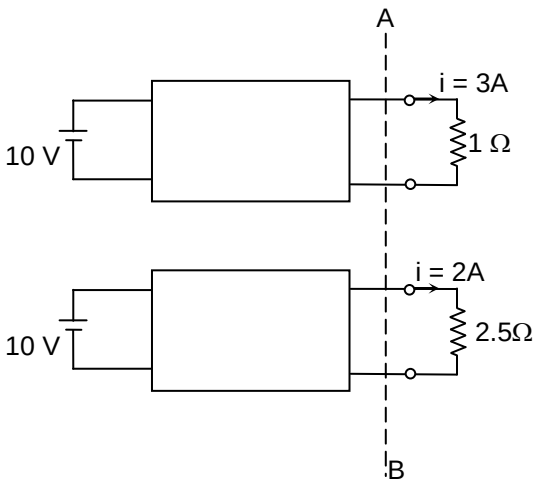
Since R = 0, i = ∞

$\therefore$ infinite current will flow for very short duration which is an impulse function.

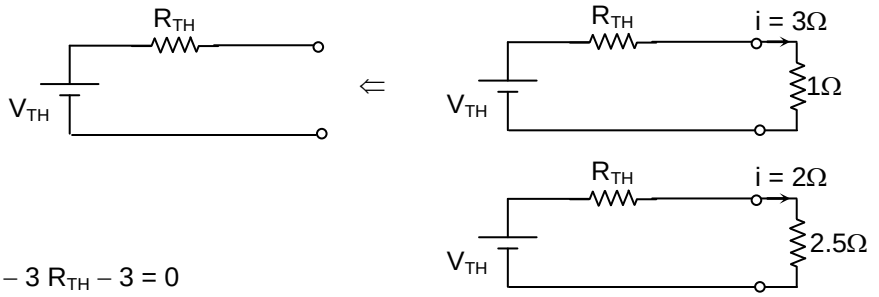
6. NA

7. (C)

Given <u>Condition 1</u>,



Now thevenizing the circuit across AB, we get



$V_{TH} - 3\,R_{TH} - 3 = 0$

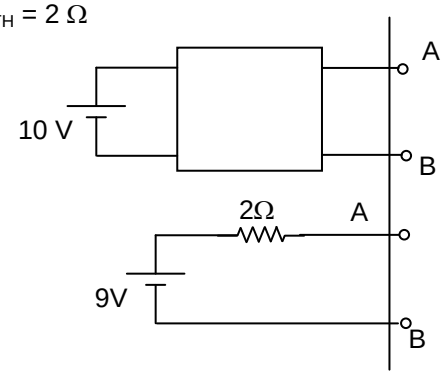
$V_{TH} - 2\,R_{TH} - 5 = 0$

∴ Solving equations simultaneously, we get

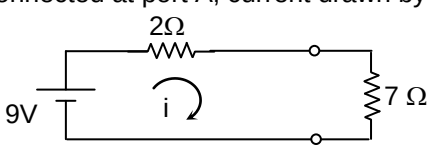
$V_{TH} = 9$ V

$R_{TH} = 2\ \Omega$

∴



With 10 V dc connected at port A, current drawn by 7Ω connected at port B is;



$\boxed{i = 1A}$

2013

1. (C)

For maximum power Transfer

$R_L = |Z_s|$

$= \sqrt{4^2 + 3^2}$

$= 5\ \Omega$

2. (C)

$V_{TH} = 10V_{L1}$

$V_{L_1} = \dfrac{V_c}{3 + j4} = \dfrac{100\angle 53.13}{5} - \text{Tan}^{-1}\left(\frac{4}{8}\right) \times j4$

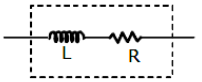
$V_{L_1} = 80\angle 90°$

$V_{TH} = 800\angle 90°$

3. (C)

Q Factor of a inductive coil.

$$\boxed{\bar{Q} = \frac{wL}{R}} \Rightarrow Q_1 = \frac{wL_1}{R_2} \ \& \ Q_2 = \frac{wL_2}{R_2}$$



When such two coils are connected in series individual inductances and resistances are added.

Hence, $L_{eq} = L_1 + L_2$

$R_{eq} = R_1 + R_2$

$$\text{Hence } Q_{eq} = \frac{\omega L_{eq}}{R_{eq}} = \frac{\omega(L_1 + L_2)}{(R_1 + R_2)} = \frac{\dfrac{\omega L_1}{R_1 R_2} + \dfrac{\omega L_2}{R_1 R_2}}{\dfrac{R_1}{R_1 R_2} + \dfrac{R_2}{R_1 R_2}}$$

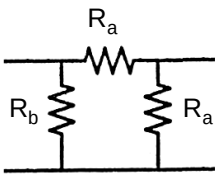
$$= \frac{\dfrac{Q_1}{R_2} + \dfrac{Q_2}{R_1}}{\dfrac{1}{R_2} + \dfrac{1}{R_1}} = \frac{Q_1 R_1 + Q_2 R_2}{R_1 + R_2}$$

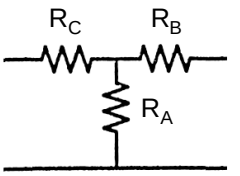
4. (B)

$$R_C = \frac{R_a R_b}{R_a + R_b + R_c}$$

$$R_B = \frac{R_a R_c}{R_a + R_b + R_c}$$

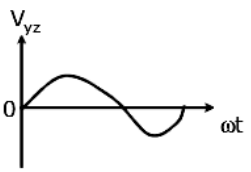
$$R_A = \frac{R_b R_c}{R_a + R_b + R_c}$$





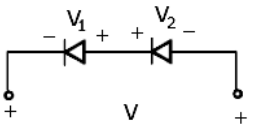
Above expression shown that if R_a, R_b & R_c is scaled by k, R_A, R_B & R_c is scaled by k only.

5. (D)



During the half cycle

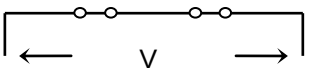
All Diodes OFF & Hence



$|V_1| = |V_2|$

V = 0V

During –ve half cycle



V = 0V.

6. (C)

$$\frac{Vc_3}{c_2+c_3} \le 5V \Rightarrow \frac{2V}{7} \le 5V \qquad ....(1)$$

$$\frac{Vc_2}{c_2+c_3} \le 2 \Rightarrow \frac{5V}{7} \le 2V \qquad ....(2)$$

$$V \le 10V \Rightarrow V \le 10V \qquad ....(3)$$

From (1), $V \le 17.5$ Volts

From (2), $V \le 2.8$ Volts

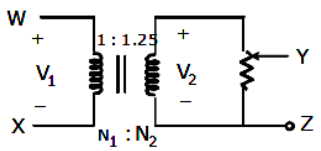
From (3), $V \le 10$ Volts

To operate Circuit safe, V should be minimum of those = 2.8V

$$c_{eff} = c_1 + (c_2 \,||\, C_3) = 10\mu F + \frac{10}{7}\mu F = \frac{80}{7}\mu F$$

$$Q = c_{eff} \times 2.8V = 32\mu c$$

7. (C)



For a transform

$$\frac{V_2}{V_1} = \frac{N_2}{N_1}$$

$\Rightarrow V_2 = 1.125 \times V_1$

The potentiometer gives an attenuation factor of 0.8 over υ_2

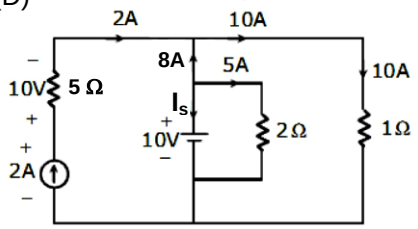
Hence $V_{yz} = 0.8\,\upsilon_2 \Rightarrow \frac{V_{yz}}{0.8} = \upsilon_2$

$$= \frac{V_{yz}}{0.8} = 1.125 \times V_{wx}$$

$$\Rightarrow V_{YZ} = V_{WX} \Rightarrow \frac{V_{YZ1}}{V_{WX1}} = \frac{100}{100}$$

Since potentiometer and transformer are bilateral elements. Hence $\frac{V_{wx2}}{V_{yz1}} = \frac{100}{100}$

8. (D)



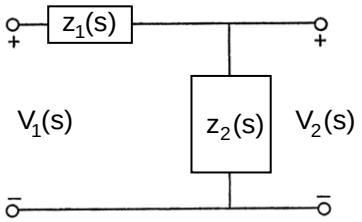
$I_s + 8 + 5 = 0$

$I_s = -13A$

9. (C)

$I_{1\Omega} = 10A$ we can use principle of superposition to determine the current across 1 ohm resistance.

10. (D)

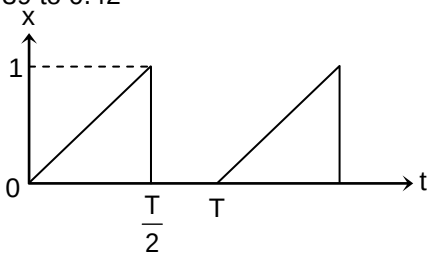


$$z_1(s) = \frac{1}{10^{-4}s'} \qquad z_2(s) = \frac{s+1}{10^{-4}s}$$

$$\frac{V_2(s)}{V_1(s)} = \frac{Z_q(s)}{z_1(s)+z_q(s)} = \frac{s+1}{s+2}$$

2014

1. 0.39 to 0.42



To find : RMS value of x

Equation of x, $x = \frac{2}{T}t$

$$X_{RMS} = \sqrt{\frac{1}{T}\int_0^T x^2(t)\,dt} = \sqrt{\frac{1}{T}\left[\int_0^{T/2}\left(\frac{2t}{T}\right)^2 dt\right]}$$

$$= \sqrt{\frac{1}{T}\left[\int_0^{T/2}\frac{4}{T^2}t^2 dt\right]} = \sqrt{\frac{1}{T}\left[\frac{4}{T^2}\frac{[t^3]_0^{T/2}}{3}\right]}$$

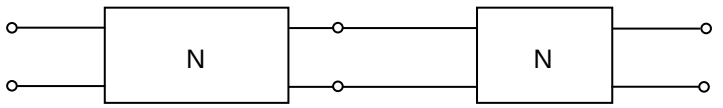
$$= \sqrt{\frac{1}{T} \times \frac{4}{T^2} \times \frac{T^3}{8} \times \frac{1}{3}} = \sqrt{\frac{1}{6}}$$

$= 0.40$

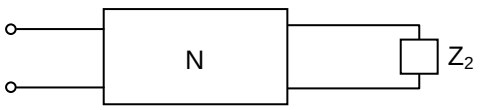
2. (C)

3. (C)

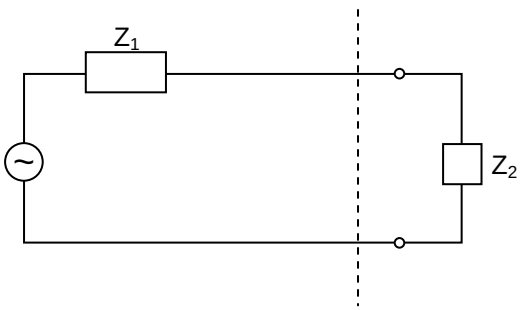
2 sections of electrical network are cascaded.



Input impedance of 2nd network acts as load of 1st network.
2nd network N can be replaced by its input impedance.



Output impedance of 1st network is Z_1.
If we replace the network by its Thevenin mode L,



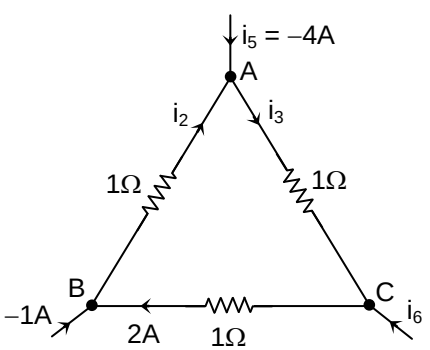
For maximum power transfer, we have

$$Z_2 = Z_1^*$$

4. (A)

Given, R = 1Ω

$i_1 = 2A$, $i_4 = -1A$, $i_5 = -4A$



Applying KCL at node B, we get

$i_2 = 2 + [-1]$

$i_2 = 1A$

Applying KCL at node A, we get

$i_2 + i_5 = i_3$

$1 + [-4] = i_3$

$\therefore \quad i_3 = -3A$

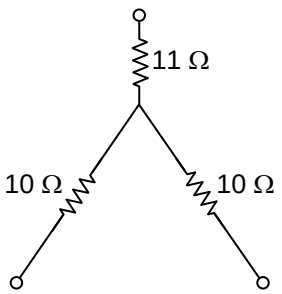
Applying KCL at node C, we get

$i_3 + i_6 = 2A$

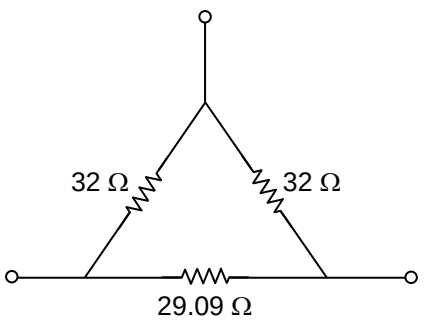
$-3 + i_6 = 2A$

$\therefore\ i_6 = 5A$

5. 29.08 to 29.10



Convert y-network to Delta-network



$\therefore$ Answer is 29.09 Ω

6. (D)

7. 0.49 to 0.51

8. 9 to 11

9. 2.79 to 2.81

10. 31.24 to 31.26

11. 9.99 to 10.01

Critically damped, $\xi = 1$

For series RLC ckt, we have

$$S^2 + \frac{R}{L}S + \frac{1}{LC} = 0$$

Comparing with $s^2 + 2\xi\omega_n s + \omega_n^2 = 0$

$$2\xi\omega_n = \frac{R}{L}$$

$$\omega_n^2 = \frac{1}{LC}$$

$$\omega_n = \frac{1}{\sqrt{LC}}$$

$$\xi = \frac{R}{2L\omega_n}$$

For critically damped we have $\xi = 1$

$$\therefore \quad 1 = \frac{R}{2L\,\omega_n}$$

$$\frac{2L \times 1}{\sqrt{LC}} = R$$

$$2\sqrt{\frac{L}{C}} = R$$

$$2\sqrt{\frac{4}{C}} = 40$$

$$\sqrt{\frac{4}{C}} = 20$$

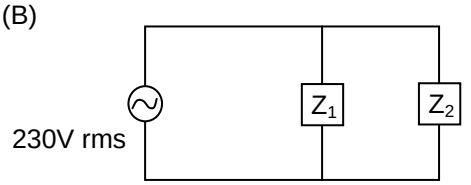
$$\frac{4}{C} = 400$$

$\therefore$ C = 0.01 F

C = 10 mF

12. 0.45 to 0.47

13. (B)



Z_1 draws 10 KW at 0.8 leading power factor.

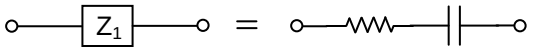
Z_2 draws 10 KVA at 0.8 lagging power factor.

In terms of complex power,

$S = S_1 + S_2$

For load Z_1, power factor is leading

$\therefore$ Z_1 is capacitive load.
Also given that Z_1 draws 10 KW



$$p.f_1 = \frac{R}{Z} = \frac{P_1}{S_1} = \cos\phi_1$$

$$0.8 = \frac{P_1}{S_1} = \frac{10\,KW}{S_1}$$

$$\therefore \quad S_1 = \frac{10\,KW}{0.8}$$

Since, load is capacitive, angle is negative.

$$\therefore \quad \bar{S}_1 = \frac{10\times10^3}{0.8}\angle-\cos^{-1}[0.8]$$

For load Z_2,
It is given that Z_2 draws 10 KVA, from the unit we come to know that given power is S_2. Since power factor is lagging, angle is positive.

$$\bar{S}_2 = 10\times10^3\angle+\cos^{-1}[0.8]$$

$$\therefore \quad \bar{S} = \bar{S}_1+\bar{S}_2$$

$$\bar{S} = \frac{10\times10^3}{0.8}\angle-\cos^{-1}[0.8]+10\times10^3\angle+\cos^{-1}[0.8]$$

$$\bar{S} = [18 - j1.5]\ KVA$$

14. 2.49 to 2.52

15. (D)

16. 1.9 to 2.1

17. (B)

18. 1.2 to 1.3

19. (A)

20. (A)

21. 1.24 to 1.26

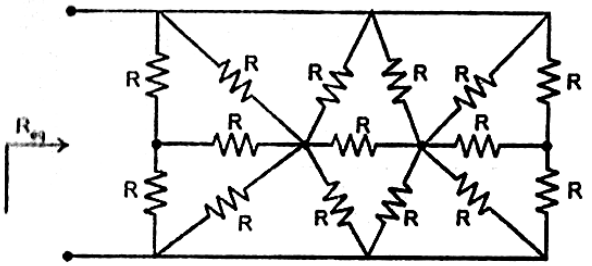
22. (B)

23. 2.60 to 2.64

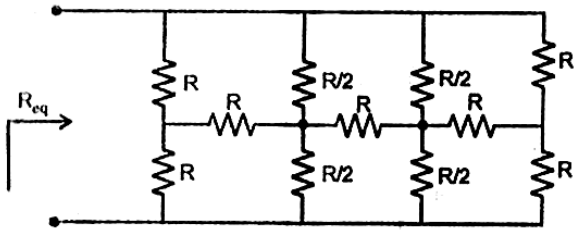
24. (C)

2015

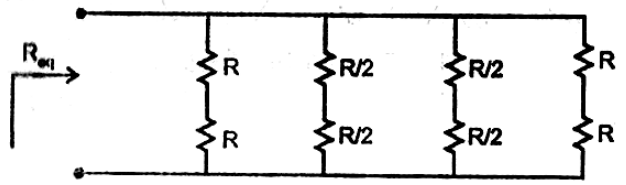
1. 99.5 to 100.5

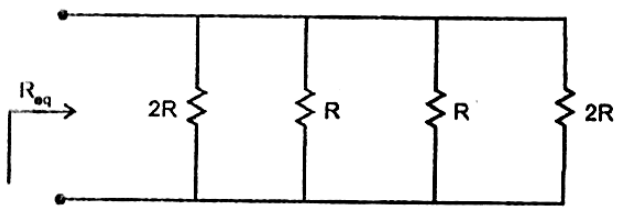


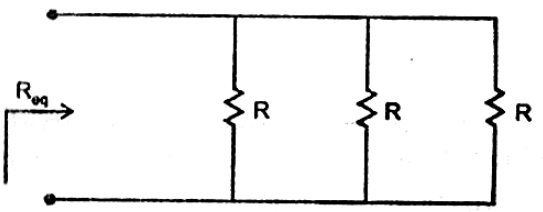
R = 300 Ω

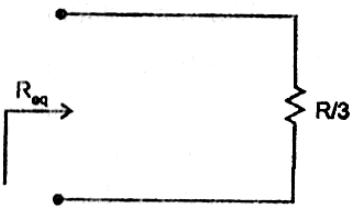


These are balanced wheatstone bridges.









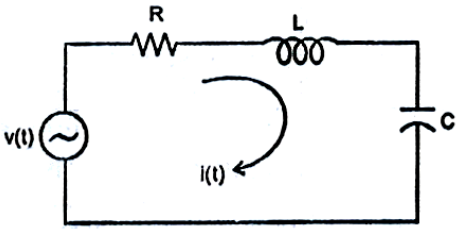
$$R_{eq} = \frac{R}{3} = \frac{300}{3} = 100\ \Omega$$

$$R_{eq} = 100\ \Omega$$

2. (C)

3. (C)

Consider series R–L–C network circuit



$$V(t) = R\,i(t) + L\frac{di(t)}{dt} + \frac{1}{C}\int i(t)\,dt$$

Taking Laplace Transformation both sides assuming initial conditions are zero.

$$V(s) = R\,I(s) + Ls\,I(s) + \frac{1}{Cs}I(s)$$

$$V(s) = \left(R + Ls + \frac{1}{Cs}\right)I(s)$$

T/F of the system

$$\frac{I(s)}{V(s)} = \frac{Cs}{(LC_s^2 + RCs + 1)}$$

$$= \frac{\frac{1}{L}s}{s^2 + \frac{R}{L}s + \frac{1}{LC}}$$

Characteristic equation of the s/s

$$s^2 + \frac{R}{L}s + \frac{1}{LC} = 0 \qquad \text{....(i)}$$

Comparing with standard characteristic equation

$$s^2 + 2\xi\omega_n s + \omega_n^2 = 0$$

$$\omega_n^2 = \frac{1}{LC}$$

$$\omega_n = \frac{1}{\sqrt{LC}}\ \text{rad/sec}$$

$$2\xi\omega_n = \frac{R}{L}$$

$$2\xi.\frac{1}{\sqrt{LC}} = \frac{R}{L}$$

$$\xi = \frac{R}{2L} \times \sqrt{LC} = \frac{R}{2}\sqrt{\frac{C}{L}}$$

Damping ratio $\xi = \frac{R}{2}\sqrt{\frac{C}{L}}$

4. 2.48 to 2.58

5. 100

$$V = \sqrt{V_R^2 + (V_c + V_L)^2}$$

$(100)^2 = (80)^2 + (V_c - 40)^2$

$(V_c - 40)^2 = (180)\ (20)$

$(V_c - 40) = \pm\sqrt{2 \times 90 \times 20}$

$V_c - 40 = \pm 60$

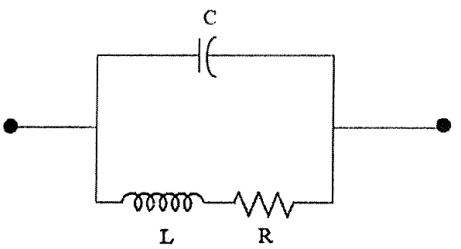
$V_c = \pm 60 + 40$

$V_c = 60 + 40$

$V_c = 100$ V

6. 4.9 to 5.1

7. (B)



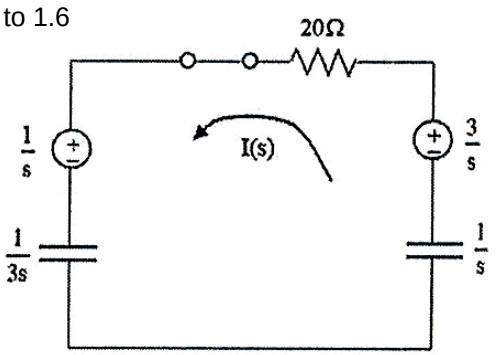
$Y = Y_c + Y_{LR}$

$$Y = j\omega C + \frac{1}{(j\omega L + R)}$$

$$= j\omega C + \frac{(R - j\omega L)}{(R^2 - \omega^2 L^2)}$$

Placing Imaginary part to zero we get option (B).

8. 1.4 to 1.6



$$I(s) = \frac{\left(\frac{3}{s} - \frac{1}{s}\right)}{\left(10 + \frac{1}{3s} + \frac{3}{3s}\right)}$$

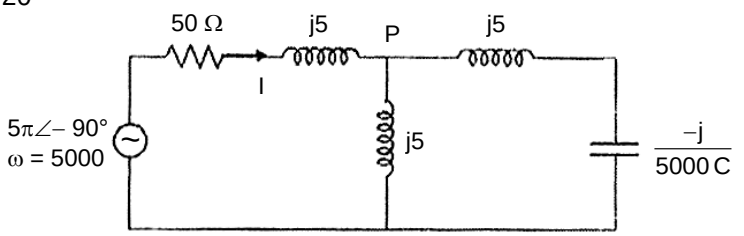
$$I(s) = \frac{2}{\left(10s + \frac{4}{3}\right)} = \frac{2}{10\left(s + \frac{4}{30}\right)}$$

$$i(t) = \frac{1}{5} e^{-\frac{4}{30}t} ; t \geq 0$$

$$E_R = \int_0^\infty i^2(t) 10\, dt = \left(\frac{10}{25}\right) \int_0^\infty e^{\frac{-4}{15}t}\, dt$$

$$= \frac{10}{25} \cdot \left. \frac{e^{\frac{4}{15}t}}{\frac{-4}{15}} \right]_0^\infty = 0 - \frac{10}{25} \times \frac{15}{-4} = 1.5 \text{ J}$$

9. 20



If I = 0, $V_p = V_i$,

$$V_i \left(\frac{1}{j5} + \frac{1}{j\left(5 - \frac{1}{5000C}\right)} \right) = 0$$

$$\Rightarrow \quad \frac{1}{j5} = \frac{-1}{j\left(5 - \frac{1}{5000C}\right)} = 0$$

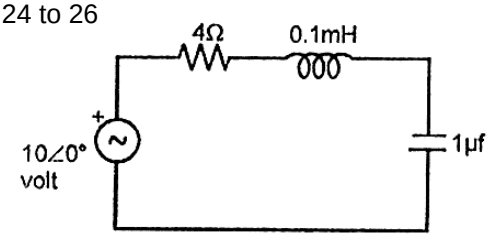
$$\frac{1}{5} = \frac{1}{\frac{1}{5000C} - 5}$$

$$\Rightarrow \quad 5 = \frac{1}{5000C} - 5$$

$$\Rightarrow \quad \frac{1}{5000C} = 10$$

$$\Rightarrow \quad C = \frac{1}{5 \times 10^4} = 20\ \mu F$$

10. 24 to 26



Resonant frequency

$$\omega_r = \frac{1}{\sqrt{LC}}$$

$$= \frac{1}{\sqrt{0.1\times10^{-3}\times1\times10^{-6}}}$$

$$= \frac{1}{\sqrt{1\times10^{-10}}} = 10^5 \text{ rad/sec}$$

$\omega_r = 10^5$ rad/sec

Current at resonance

$$I = \frac{V}{R} = \frac{10}{4} = 2.5 \text{ Ap}$$

$X_L = W_L = 10^5 \times 0.1 \times 10^{-3} = 10\ \Omega$

$V_L = IX_L = 2.5 \times 10$

$= 25$ Volt

At Resonance

$V_C = V_L$

$V_C = 25$ volt

11. (A)

By nodal analysis

$$-5 + I + I + 2I = 0$$

$$4I = 5$$

$$I = \frac{5}{4} A$$

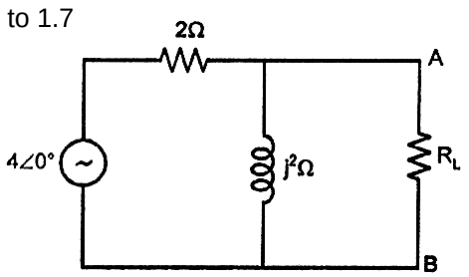
$V_1 = 4I = 5$ volts

$V_2 = 4(5) + V_1$

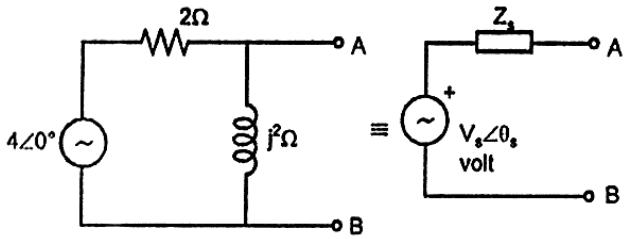
$= 20 + V_1$

$= 25$ volts

12. 1.6 to 1.7



Thevenins equivalent



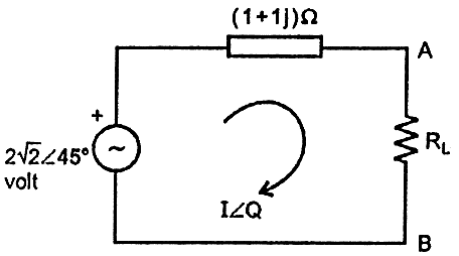
$$V_S \angle \theta_S = 4\angle 0° \times \frac{j2}{(2+j2)}$$

$$= 2.828 \angle 45° \text{ volt}$$

$$= 2\sqrt{2} \angle 45° \text{ volt}$$

$$z_S = \frac{2\times(j2)}{(2+j2)}$$

$$= (1 + 1j)\Omega$$



$$I \angle\theta = \frac{2\sqrt{2}\angle 45°}{(1+R_L)+j1}$$

$$P_L = I^2 R_L$$

where $\quad I = |I \angle \theta|$

$$= \frac{2\sqrt{2}}{\sqrt{(1+R_L)^2+(1)^2}}$$

$$P_L = \frac{8.R_L}{\left\{(R_L+1)^2+1\right\}}$$

For maximum power transfer to the load

$$\frac{dP_L}{dR_L} = 0$$

$$\frac{dP_L}{dR_L} = 8\left\{\frac{\left((R_L+1)^2+1\right).1 - R_L\left(2(R_L+1)\right)}{\left\{(R_L+1)^2+1\right\}^2}\right\} = 0$$

$$= 8\{(R_L + 1)^2 + 1 - 2R_L^2 - 2R_L\} = 0$$

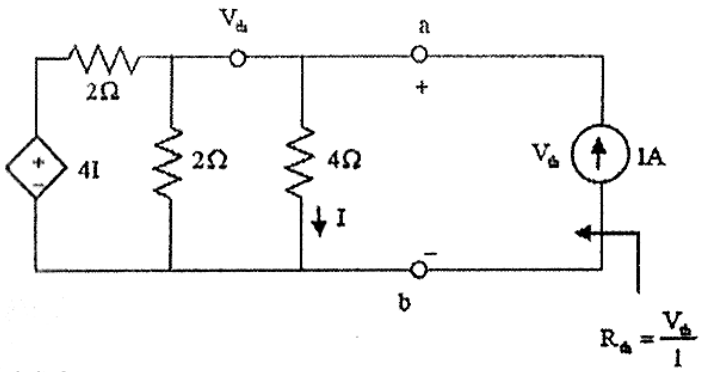
$$8\{2 - R_L^2\} = 0$$

$$R_L = \sqrt{2}\,\Omega$$

$$P_L = \frac{8}{\left(\sqrt{2}+1\right)^2+1} \times \sqrt{2}$$

P_L = 1.66 watts

13. 1.30 to 1.35



Nodal @ 'a'

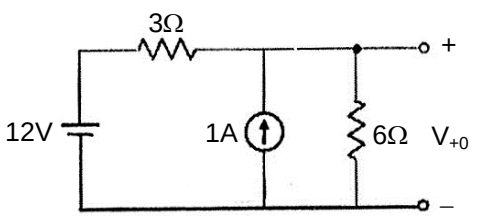
$$\frac{V_{th}-4I}{2}+\frac{V_{th}}{2}+\frac{V_{th}}{4}-1=0$$

Also $V_{th} = 4I$

$\Rightarrow$ $V_{th} = \frac{4}{3}$

$R_{th} = \frac{4}{3} = 1\frac{1}{3} = 1.333$

14. 10



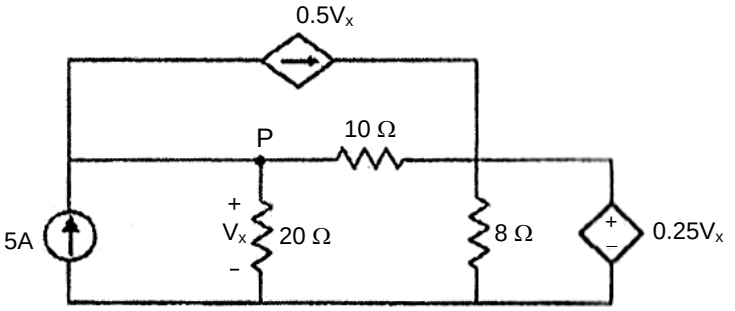
Applying nodal equation at point P

$$V_{+n}\left(\frac{1}{3}+\frac{1}{6}\right) = \frac{12}{3}+1$$

$$\frac{V_{+n}}{2} = 5$$

$\Rightarrow$ V_{+n} = 10 V

15. 7.95 to 8.05



Apply KCL at point P

$$\frac{V_x}{20} + \frac{V_x - 0.25\,V_x}{10} + 0.5\,V_x = 5$$

$$V_x\left(\frac{1}{20} + \frac{0.75}{10} + 0.5\right) = 5$$

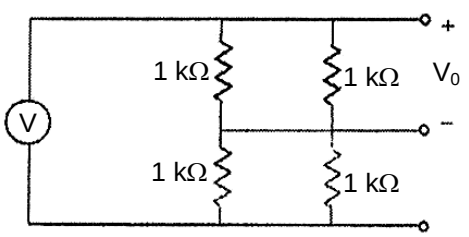
$$V_x\left(\frac{5}{8}\right) = 5$$

$\Rightarrow \quad V_x = 8V$

16. 0.49 to 0.51

For capacitor $Z_C = \dfrac{1}{j\omega C}$

Very high frequency means $\omega \to \infty \Rightarrow Z_C \to 0$
So, all capacitors are replaced by short circuit.



By voltage division, $V_0 = \dfrac{V_i}{2}$

$$V_0 = \frac{1.\sin\omega t}{2} = 0.5 \sin \omega t$$

Thus, Peak voltage = 0.5

17. (A)

18. (B)

For the standard 'T' network, obtain the Z-matrix first and then convert it into T-Matrix

$$Z = \begin{bmatrix} 7 + j4 & 2 \\ 2 & 7 - j4 \end{bmatrix}$$

$\Delta Z = [(7 + j4)\,(7 - j4)] - 4 = 49 + 16 - 4 = 16$

$$A = \frac{Z_{11}}{Z_{21}} = \frac{7+j4}{2} = 3.5 + j2$$

$$B = \frac{\Delta Z}{Z_{21}} = \frac{61}{2} = 30.5$$

$$C = \frac{1}{Z_{21}} = \frac{1}{2} = 0.5$$

$$D = \frac{Z_{22}}{Z_{21}} = \frac{7-j4}{2} = 3.5 - j2$$

$$T = \begin{bmatrix} 3.5+j2 & 30.5 \\ 0.5 & 3.5-j2 \end{bmatrix}$$

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1. (A)

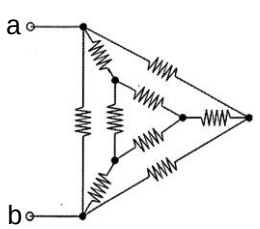
The property of any LTI system or network is if the excitation contains 'n' number of different frequency then the response also contains exactly n number of different frequency term and the output frequency and input frequency must be same however depending on components there is a possible change in amplitude and phase but never the frequency.

→ If the source has 3 frequency terms as given $\sum_{k=1}^{3} a_k \cos k\omega_0 t$ then any voltage or any current of any element should have also 3 terms based on this option (B) and (D) are eliminated.

→ If we take option (C). It has 3 frequency term but it also suggest there is a phase change so ϕ_k but amplitude must be same as input as a_k is present which may not be true always.

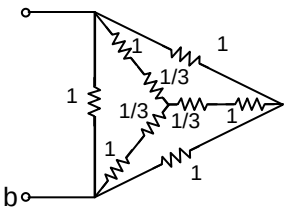
→ So option (A) is correct, as it suggest frequency term of output and inputs are same with possible change in amplitude and phase, because we have (b_k and ϕ_k).

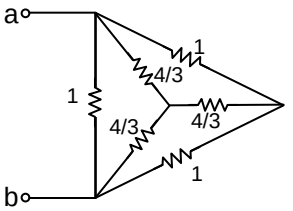
2. (D)



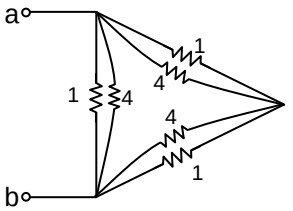
Each resistor has value 1

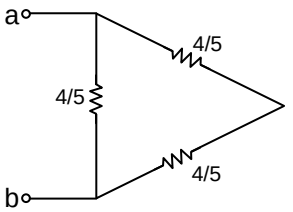
We do delta to star conversion of inner delta.





Converting inner star to delta



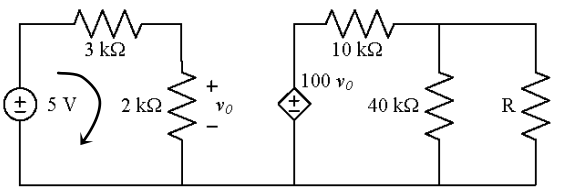


$\therefore$ Req = $\frac{4}{5} /\!/ \frac{8}{5}$

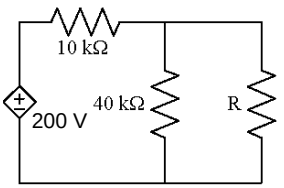
$$= \frac{\frac{4}{5} \times \frac{8}{5}}{\frac{4}{5} + \frac{8}{5}} = \frac{\frac{32}{25}}{\frac{12}{5}} = \frac{32}{25} \times \frac{5}{12}$$

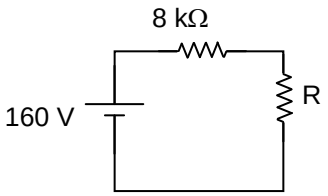
Req = $\frac{8}{15}$ = 0.54 Ω

3. 0.78 to 0.82



$v_0 = \frac{5 \times 2}{2 + 3}$ = 2V





$$P_{max} = \frac{Vth^2}{4Rth} = \frac{(160)^2}{4 \times 8 \times 10^3} = 0.8 \text{ W}$$

4. (B)

For I_L to absorb power, V_L should be positive.

$\therefore$ Power absorbed by constant current source $[I_L] = V_L I_L$

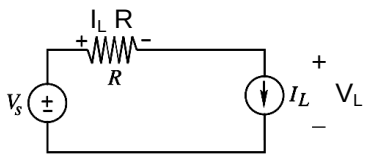
By Kvl, we get

$V_L = V_S - I_L R$

$\therefore$ $P_L = V_L I_L$

$P_L = [V_S - I_L R] I_L$

$= V_S I_L - I_L^2 R$



For P_L to be maximum,

$$\frac{dP_L}{dI_L} = 0$$

$V_S - 2I_L R = 0$

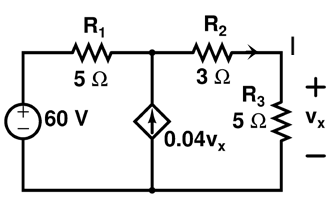
$$\therefore \quad I_L = \frac{V_S}{2R}$$

Hence, for P_L to be maximum, I_L should be $\frac{V_S}{2R}$

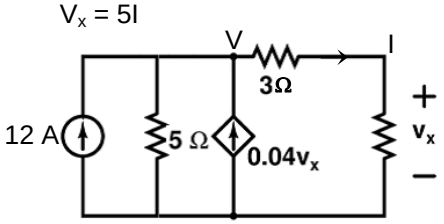
$$\therefore \quad \text{Maximum Power absorbed is} = V_S I_L - I_L^2 R = V_S \left[\frac{V_S}{2R}\right] - \left[\frac{V_S}{2R}\right]^2 R = \frac{V_S^2}{2R} - \frac{V_S^2}{4R}$$

$$P_L = \frac{V_S^2}{4R}$$

5. 4.9 to 5.1



$V_x = 5I$



$12 + 0.04V_x = \frac{V}{5} + I$

$V = V_x + 3I$

$\therefore \quad 12 + 0.04V_x = \frac{[V_x + 3I]}{5} + I$

$12 + 0.04\,V_x = \frac{V_x}{5} + \frac{3}{5}I + I$

$12 + 0.04\,V_x = 0.2V_x + \frac{8}{5}\,I$

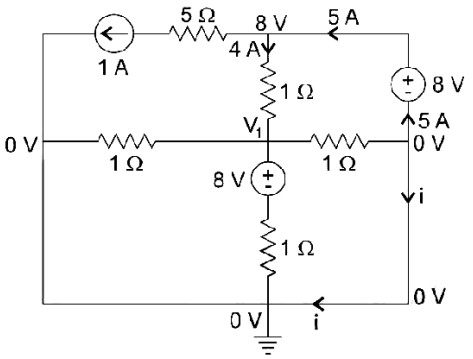
$12 = 0.16\,V_x + \frac{8}{5}\,I$

$12 = 0.16[5I] + \frac{8}{5}\,I$

$12 = 2.4\,I$

$\therefore \quad I = 5A$

6. –1.05 to –0.95



Apply Nodal we get

$\frac{V_1 - 0}{1} + \frac{V_1 - 8}{1} + \frac{V_1 - 0}{1} + \frac{V_1 - 8}{1} = 0$

$4V_1 - 16 = 0 \Rightarrow V_1 = 4V$

Again Nodal analysis gives

$i + \frac{(0 - V_1)}{1} + 5 = 0 \Rightarrow i = V_1 - 5 = 4 - 5 = -1$ Amp

7. 0.30 to 0.34

At resonance, $\frac{|I_L|}{|I_R|}$ = Quality factor

For Parallel RLC ckt

$Q = R\sqrt{\frac{C}{L}}$

$Q = 10\sqrt{\frac{10 \times 10^{-6}}{10 \times 10^{-3}}}$

$Q = 0.3162$

8. (B)

$v_i(t) = 2\cos(200t) + 4\sin(500\,t)$ given

we can use superposition theorem

Case (1)

when input is 2 cos (200t)

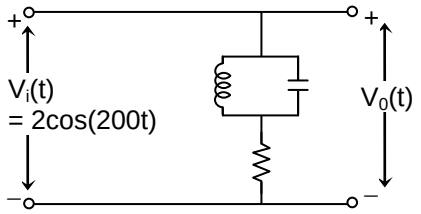
w = 200 rad/sec

At w = 200 rad/sec,

$jx_L = jw_L = j \times 200 \times 0.25 = j50$

$-jx_L = -j/wc = -j/200 \times 100 \times 10^{-6} = -j50$

∴ At w = 200 rad/sec, series combination of L = 0.25H and C = 100μF acts like a short circuit



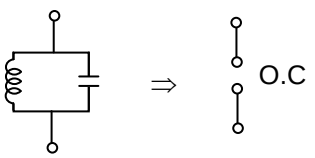
∴ $V_0(t) = 2\cos(200t)$

Case (2)

When input is 4 sin (500t)

w = 500 rad/sec

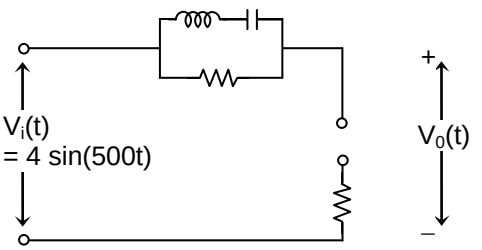
At w = 500 rad/sec



$jx_L = jwL = j \times 500 \times 0.4 = j200$

$-jx_L = -j/wc = -j/500 \times 10 \times 10^{-6} = -j\,200$

∴ At w = 500 rad/sec, Parallel combination of L = 0.4H and C = 10μF acts like an open circuit.



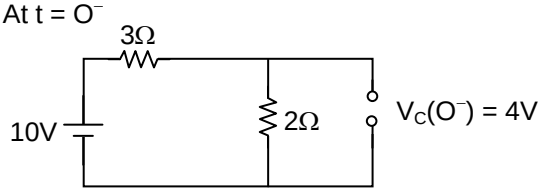
∴ $V_0(t) = 4\sin(500\,t)$

∴ By superposition,

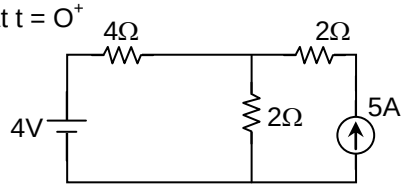
Total $V_0(t) = 2\cos(200t) + 4\sin(500t)$

9. (D)

At $t = O^-$

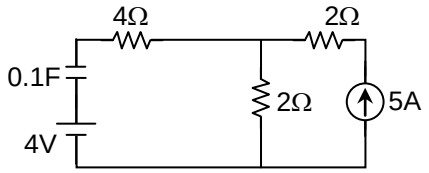


At t = O^+

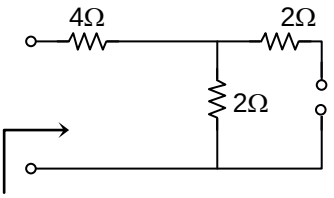


$\therefore \quad V_C(O^+) = 4V$

For t > 0,



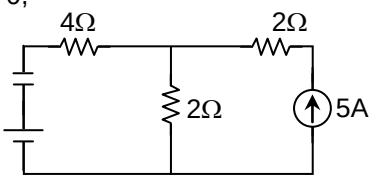
To find time constant of capacitor $\rightarrow$ remove capacitor
$\rightarrow$ open ckt current source
$\rightarrow$ find equivalent resistance



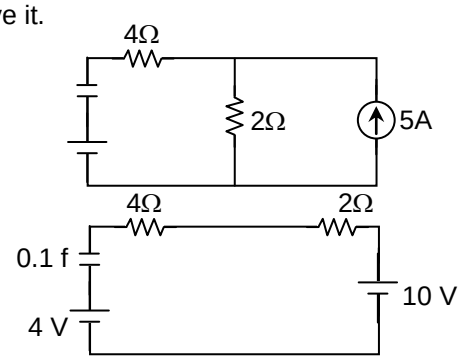
Req = 6Ω

$\therefore \quad Z = Req\ C \quad = 6 \times 0.1$
$= 0.6$

For t > 0,



2Ω resister in series with current source [5A] is redundant. Therefore, we can remove it.



$\therefore$ After switch position is charged, capacitor will be charged by 10V, source

$\therefore \quad V_C(\infty) = 10$ V

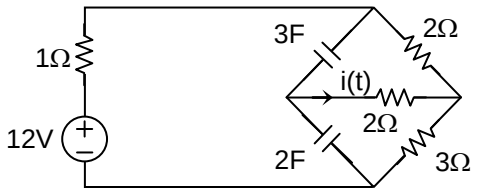
$\therefore \quad V_C(t) = V_C(\infty) - [V_C(\infty) - V_C(0^+)]\, e^{-t/\tau}$

$\therefore \quad V_C(t) = 10 - [10 - 4]\, e^{-t/0.6}$

$V_C(t) = 10 - 6e^{-t/0.6}$ V

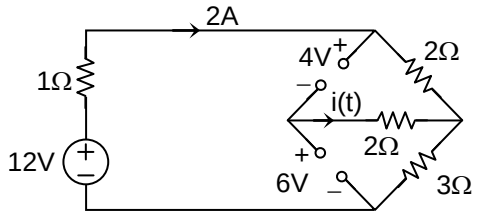
10. 99 to 101

11. 0.98 to 1.02



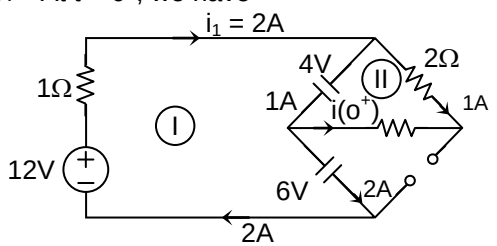
At t < 0, circuit is in steady state

$\therefore$ Capacitor open circuit.



At t = 0, 3 Ω resistor burns out and becomes open circuit

$\therefore$ At $t = 0^+$, we have



Apply Kvl in loop (I), we get

$12 - 1(i_1) - 4 - 6 = 0$

$i_1 = 2A$

Applying Kvl in loop (II), we get

$2[i(0^+)] + 2[i(0^+)] + 4 = 0$

$4\, i(0^+) + 4 = 0$

$i(0^+) = -1A$

Since, in the question current direction is not mentioned and it is not clear whether current in 2Ω resister is asked or current in 2F capacitor.

$\therefore$ Correct answer are $\rightarrow$ 1A, –1A, +2A, –2A.

12. (D)

13. 2.98 to 3.02

$$Z = \begin{bmatrix} 2j\omega & j\omega \\ j\omega & 3 + 2j\omega \end{bmatrix}$$

$V_1 = Z_{11}\, I_1 + Z_{12}\, I_2$

$V_2 = Z_{21}\, I_1 + Z_{22}\, I_2$

$$Z_{11} = \left.\frac{V_1}{I_1}\right|_{I_2=0}$$

$Z_{11} = Za + Zc = 2j\omega$

$Z_{22} = Zb + Zc = 3 + 2j\omega$

$$Z_{12} = \left.\frac{V_1}{I_2}\right|_{I_1=0} = Z_C$$

$\therefore \quad Z_C = j\omega$

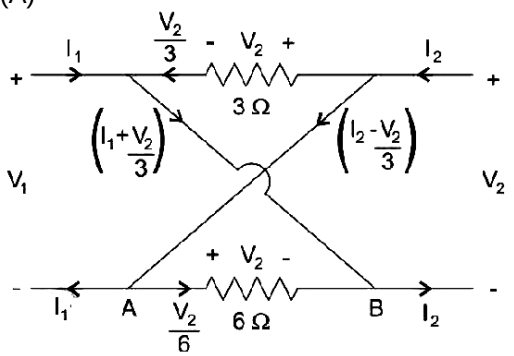
Hence, $Zb + Zc = 3 + 2j\omega$

$Zb = 3 + 2j\omega - j\omega$

$Zb = 3 + j\omega$

$\therefore \quad Rb = 3\ \Omega$

14. (A)



Apply Nodal at A, we get $\quad I_2 - \frac{V_2}{3} = I_1 + \frac{V_2}{6}$

$$I_2 = I_1 + \frac{V_2}{2} \qquad \ldots(1)$$

$I_1 = 0 \Rightarrow V_2 = 2I_2 \Rightarrow Z_{22} = 2$

$I_2 = 0 \Rightarrow V_2 = -2I_1 \Rightarrow Z_{21} = -2$

And form the figure we get $V_1 = -V_2 \Rightarrow Z = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}$

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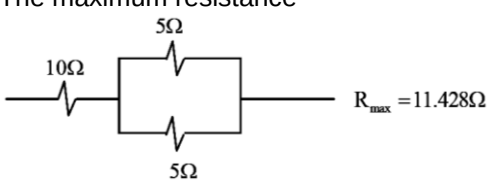
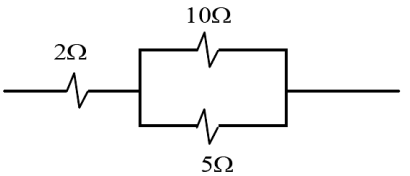
1. 2.12 to 2.16

The maximum resistance



The minimum resistance



$R_{min} = 5.333\Omega$

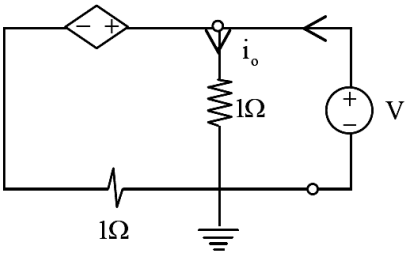
$$\frac{R_{max}}{R_{min}} = 2.14$$

2. 3.15 to 3.21

$$V_0 = \frac{V_m}{\pi} = \frac{10}{\pi} = 3.1847V$$

3. –1.01 to –0.99

To find $R_{th} = \frac{V}{I}$



Here $i_0 = \frac{V}{1\Omega} = V$

Nodal at V

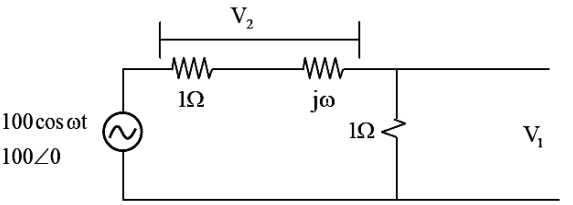
$$\frac{V - 3i_0}{I} + \frac{V}{I} - I = 0$$

$V - 3V + V - I = 0$

$R_{th} = -1\Omega$

4. 0.9 to 1.1

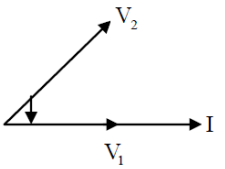
$$V_1 = \left(\frac{1}{2 + j\omega}\right) \times 1\omega\angle 0$$



$$V_1 = \frac{100}{\sqrt{4 + \omega^2}} \angle -\tan^{-1}\omega / 2$$

$$V_2 = \frac{1 + j\omega}{2 + j\omega} 100\angle 0$$

$$V_2 = \left(\frac{\sqrt{1 + \omega^2}}{\sqrt{4 + \omega^2}}\right) \times 1\omega\angle -\tan^{-1}\frac{\omega}{2} + \tan^{-1}\omega$$



$$V_2 - V_1 = \pi / 4$$

$$-\tan^{-1}\frac{\omega}{2} + \tan^{-1}\omega + \tan^{-1}\frac{\omega}{2} = \pi / 4$$

$$\omega = \tan\pi/4 = 1 \text{ rad / sec}$$

5. 2.55 to 2.65

$$V_1 = \left(\frac{4 + j3}{4 + j3 + 5 - 12j}\right) \times 100\angle 0 \Rightarrow V_1 = \left(\frac{4 + j3}{9 - 9j}\right) \times 100\angle 0$$

$$V_2 = \left(\frac{5 - 12j}{4 + j3 + 5 - 12j}\right) \times 100\angle 0 \Rightarrow V_2 = \left(\frac{5 - 12j}{9 - 9j}\right) \times 100\angle 0$$

$$\left|\frac{V_2}{V_1}\right| = \left|\frac{5-12j}{4+j3}\right| = \frac{\sqrt{5^2+12^2}}{\sqrt{4^2+3^2}} = \frac{13}{5} = 2.6$$

6. 0.30 to 0.40

Under dc condition inductor acts as short all

$\therefore \quad I_{total} = \frac{15}{1} = 15A$

$i(t)=(i(0^-) - i(\infty))e^{-t/2} + i(\infty)$

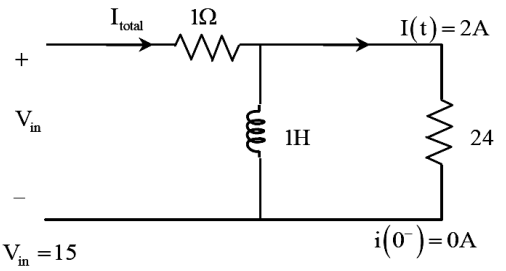
$i(0^-) = i(0^+) = 0A$

$i(t) = (0-15)e^{-3/2t} + 15$

$i_{total}(t) = 15\left(1-e^{-3/2t}\right)A$

$i(t)_{total} = \frac{I_{total}}{3} = \frac{15}{3}\left(1-e^{-3/2t}\right)$

$2 = 5\left(1-e^{-3/2t}\right) \Rightarrow t = 0.34$ sec



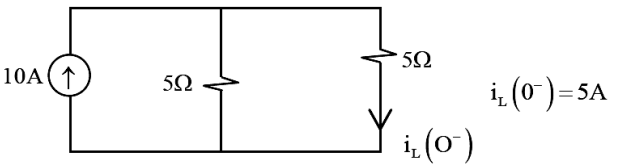
7. 0.19 to 0.21

If I & V are in phase then the circuit is in resonance at resonance

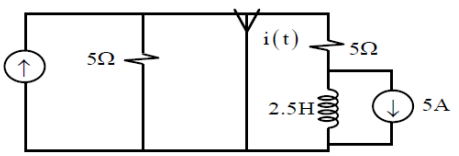
$$\left|\frac{V_C}{V_R}\right| = Q = \frac{1}{R}\sqrt{\frac{L}{C}} = \frac{1}{5}\sqrt{\frac{5}{5}} = 0.2$$

8. 8.0 to 8.3

At $t = 0^-$



At $t \geq 0$



$\tau = \frac{L}{R} = \frac{2.5}{5} = \frac{1}{2} \Rightarrow i_L(\infty) = 0$

$I_L = 5e^{-2t}$

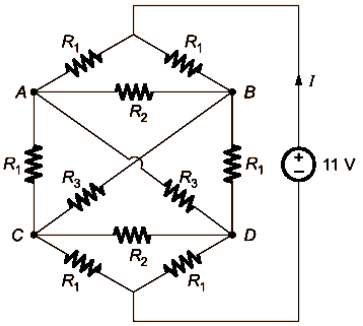
$i(t) = 10 - 5e^{-2t}$

At t = 0.5s

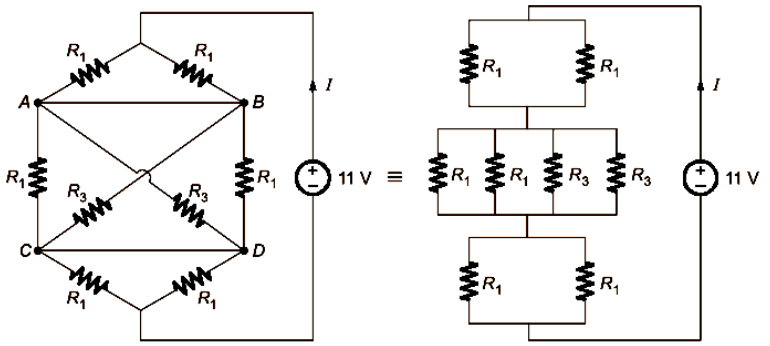
$i(0.5) = 10 - \frac{5}{e} = 8.16A$

2018

1. 7.9 to 8.1



As the network is symmetric,
$V_A = V_B$ and V_C and V_D
So, current through R_2 resistors is zero and as $V_A = V_B$ and $V_C = V_D$, electrically the circuit can be reduced as.



Total resistance, $R_T = 2(R_1 \| R_1) + (R_1 \| R_1 \| R_3 \| R_3) = R_1 + \left(\frac{R_1}{2} \left\| \frac{R_3}{2}\right.\right)$

Given that, $R_1 = 1\ \Omega$ and $R_3 = 3\ \Omega$

So, $R_T = 1 + \left(\frac{1}{2} \left\| \frac{3}{2}\right.\right)\Omega = 1 + \frac{3/2}{4} = \frac{11}{8}\Omega$

$$I = \frac{11V}{R_T} = \frac{11}{(11/8)} = 8\text{ A}$$

2. 0.284 to 0.348

Loop current, $i(t) = \frac{1}{1+1}\left(1 - e^{-t/\tau}\right)A;\ t \ge 0$

$$\tau = \frac{L}{R_{eq}} = \frac{1}{1+1} = \frac{1}{2}\text{sec}$$

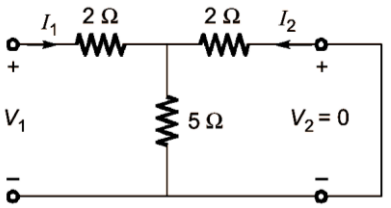
$$i(t) = \frac{1}{2}\left(1 - e^{-2t}\right)A;\ t \ge 0$$

At t = 0.5 sec, $i(t) = \frac{1}{2}\left(1 - e^{-1}\right)A = 0.316A$

3. 4.3 to 5.3

$$B = -\left.\frac{V_1}{I_2}\right|_{V_2=0}$$

When $V_2 = 0$ (i.e. when port–2 is short circuited),



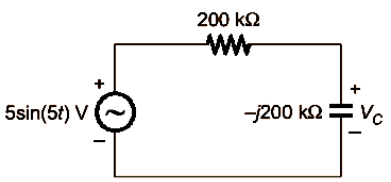
$$I_1 = \frac{V_1}{2\Omega + (5\Omega \| 2\Omega)} = \frac{7V_1}{24\Omega}$$

$$I_2 = -I_1 \times \frac{5\Omega}{5\Omega + 2\Omega} = \frac{-5V_1}{24\Omega}$$

So, $B = -\frac{V_1}{I_2} = \frac{24}{5}\Omega = 4.80\Omega$

4. (C)

$$\frac{1}{\omega C} = \frac{1}{5 \times 10^{-6}} = 200\,k\Omega$$



$$V_c = \frac{5\angle 0°}{200 - j200} \times (-j200)V = \frac{5 \le 0° \times 1\angle -90°}{\sqrt{2}\angle -45°}$$

$$= \frac{5}{2}\angle -45° \text{ V} = 2.5\sqrt{2}\sin\left(5t - \frac{\pi}{4}\right)V$$

$$= 2.5\sqrt{2}\sin(5t - 0.25\pi)V$$

2019

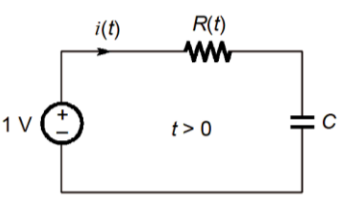
1. 0.23 to 0.27

$$T = 3R_0C = 3\text{sec}$$

$$R(t) = \left(1 - \frac{t}{3}\right); 0 \le t \le 3\text{ sec}$$

$$R(t)i(t) + \frac{1}{C}\int i(t)\,dt = 1$$

$$\left(1 - \frac{1}{3}\right)i(t) + \int i(t)\,dt = 1$$



Differentiating both sides, we get,

$$\left(1-\frac{t}{3}\right)\frac{di}{dt}-\frac{i}{3}+i=0$$

$$(3-t)\frac{di}{dt}+2i=0$$

$$\frac{di}{i}=-\frac{2}{(3-t)}dt$$

Integrating on both sides, we get,

$\ln(i) = 2\ln(3-t) + \ln(c)$

$i(t) = c(3-t)^2;\ t \geq 0$

Given that, $i(0) = 1A$

So, $c(3-0)^2 = 1A$

$$C = \frac{1}{9}A$$

$$i(t)=\frac{1}{9}(3-t)^2\ A$$

At $t = \frac{T}{2}=1.5\,\text{sec}$ $\quad i(1.5) = \frac{1}{9}(1.5)^2=0.25A$

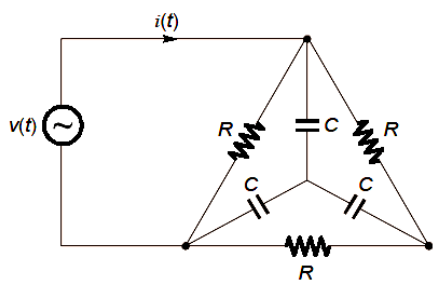
2. (B)

According to reciprocity theorem,

In a linear bilateral single source network the ratio of response to excitation remains the same even after their positions get interchanged.

$$\therefore\quad \frac{I}{5}=\frac{1}{5}\Rightarrow I=1A$$

3. (C)



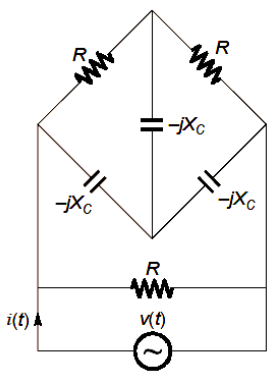
Here, $X_c=\frac{1}{\omega C}=\frac{1}{10^3\times10^{-6}}=\frac{1}{10^{-3}}$

$X_c = 10^3\ \Omega$

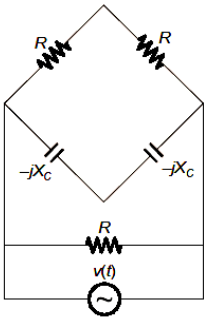
$R = 10^3\ \Omega$ (Given)

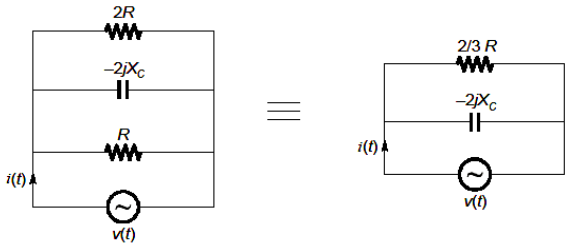
$v(t) = 2\sin 1000t\ V = 2\angle 0°\ V$

Redrawing the given network, we get,



A the bridge is balanced, it can be redrawn as





$$\therefore \quad Y_{eq} = Y_1 + Y_2 = \frac{3}{2}R + \frac{1}{-2jX_c} = \frac{3}{2}\times 10^{-3} + j\frac{1}{2}\times 10^{-3}$$

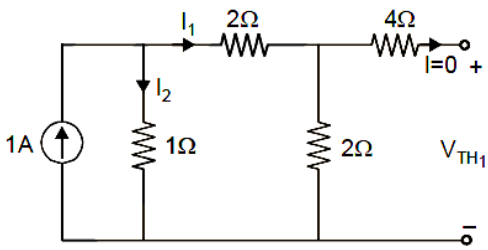
$$\therefore \quad i(t) = v(t)\times Y_{eq} = 2\angle 0^0\left[\frac{3}{2} + j\frac{1}{2}\right]mA$$

= (3 + j 1) mA = 3 sin (1000t) + cos (1000t) mA

2020

1. (C)

Case(i) : Only current source, 1A is there and voltage source, 2V and current source, 2A are short circuited and open circuited respectively.

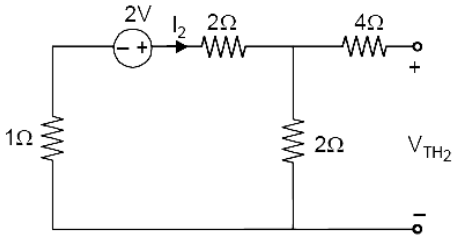


From current division rule,

$$I_1 = \frac{1}{4+1}\times 1 = \frac{1}{5}A$$

$$\therefore \quad V_{TH1} = 2\times I_1 = 2\times\frac{1}{5} = \frac{2}{5}A$$

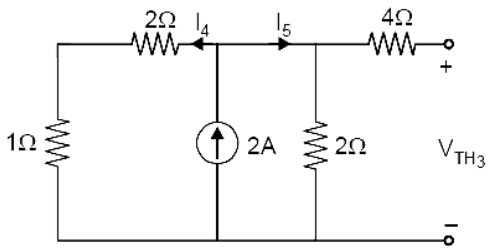
Case II: Only voltage source 2V is there and both current source 1A and 2A are open circuited.



From Circuit, $I_3 = \dfrac{2}{2+2+1} = \dfrac{2}{5}A$

$V_{TH_2} = \dfrac{2}{5} \times 2 = \dfrac{4}{5}V$

Case III: Only current source, 2A is there and voltage source 2V and current source 1A are short circuited and open circuited respectively.



From current division rule,

$I_5 = \dfrac{(1+2)}{(1+2+2)} \times 2 = \dfrac{6}{5}A$

$\therefore\ V_{TH_3} = 2 \times I_5 = \dfrac{12}{5}A$

Superposition theorem,

$V_{TH} = V_{TH_1} + V_{TH_2} + V_{TH_3}$

$= \dfrac{2}{5} + \dfrac{4}{5} + \dfrac{12}{5} = \dfrac{18}{5} = 3.6V$

2. 48

$[Z] = \begin{bmatrix} 40 & 60 \\ 60 & 120 \end{bmatrix}$

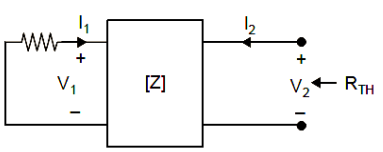
$\therefore\ V_1 = 40\ I_1 + 60\ I_2$ ….(i)

$V_2 = 60\ I_1 + 120\ I_2$ ….(ii)

For maximum power transfer,

$Z_L = R_{TH}$

To find the R_{TH}, the given circuit is redrawn as:



From the above circuit,

$$V_1 = -10\ I_1 \text{ and } R_{TH} = \frac{V_2}{I_2}$$

Putting $V_1 = 10\ I_1$ in eq. (i), we get

$$-10\ I_1 = 40\ I_1 + 60\ I_2$$

$$\Rightarrow\ -5\ I_1 = 6\ I_2 \Rightarrow I_1 = \frac{-5}{5} I_2$$

Putting $I_1 = \frac{-6}{5} I_2$ in eq. (ii), we get

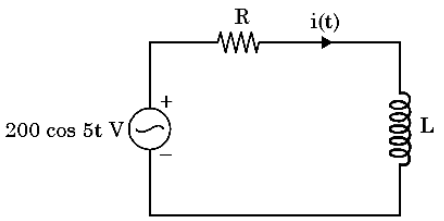
$$\Rightarrow\ V_2 = 60\left(\frac{-6}{5}\right) I_2 + 120 I_2$$

$$\Rightarrow\ V_2 = -12 I_2 + 120 I_2 = 45 I_2$$

$$\Rightarrow\ \frac{V_2}{I_2} = R_{TH} = 48\,\Omega$$

3. 2.80 to 2.85

Given,



$$V(t) = 200 \cos 5t \text{ V}$$

$$= \frac{200}{\sqrt{2}} \angle 0^\circ \text{ V}$$

$$i(t) = 10 \cos\left(5t - \frac{\pi}{4}\right) \text{A}$$

$$= \frac{10}{\sqrt{2}} \angle \frac{-\pi}{4} \text{A}$$

Here, $\omega = 5$ rad/sec

Impedance of the circuit,

$$z = \frac{V(t)}{i(t)} = \frac{200/\sqrt{2}\ \angle 0^\circ}{10/\sqrt{2}\ \angle \frac{-\pi}{4}}$$

$$= 20 \angle \frac{\pi}{4}$$

$$= 20\left(\cos\frac{\pi}{4} + j \sin\frac{\pi}{4}\right)$$

$$= \frac{20}{\sqrt{2}} + j\frac{20}{\sqrt{2}}$$

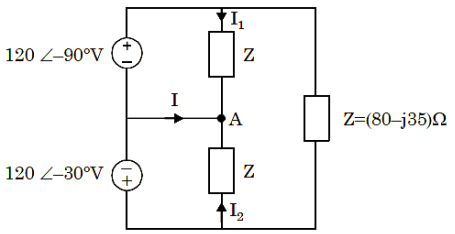
$$= R + jX_L$$

$$\therefore \quad X_L = \frac{20}{\sqrt{2}}$$

$$\Rightarrow \quad \omega L = \frac{20}{\sqrt{2}}$$

$$\Rightarrow \quad L = \frac{20}{\sqrt{2} \times 5} = \frac{4}{\sqrt{2}} = 2.83\,H$$

4. (C)



From the circuit,

$$I_1 = \frac{120\angle -90°}{Z} = \frac{120\angle -90°}{80 - j35}$$

$$= 1.374\angle -66.37°A$$

$$I_2 = \frac{120\angle -30°}{Z} = \frac{120\angle -30°}{80 - j35}$$

$$= 1.374\angle -6.37°A$$

Applying KCL at node A, we get:

$I + I_1 + I_2 = 0$

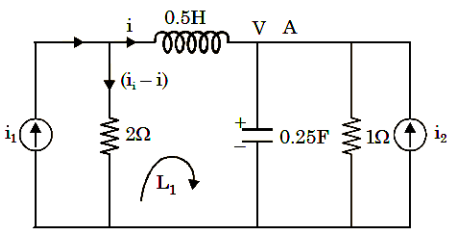
$\Rightarrow \quad I = -(I_1 + I_2)$

$= -(1.374 \angle -66.37° + 1.374° \angle -6.37°)$

$= 2.38 \angle 143.63°A$

5. (A)

Given figure,



Applying KVL in loop L_1

$$\Rightarrow \quad -2(i_1 - i) + 0.5\frac{di}{dt} + V = 0$$

$$\Rightarrow \quad -2i_1 + 2i + 0.5\frac{di}{dt} + V = 0$$

$$\Rightarrow \quad \frac{di}{dt} = [2i_1 - 2i - V]\frac{1}{0.5} = 4i_1 - 4i - 2V$$

$$\Rightarrow \quad \frac{di}{dt} = -2V - 4i + 4i_1 \qquad \text{....(i)}$$

Applying KVL at node A

$$\Rightarrow \quad I_2 + \frac{(0-V)}{(1)} + i = \left(\frac{\partial V}{\partial t}\right)(0.25)$$

$$\Rightarrow \quad 4[I_2 - V + i] = \frac{\partial V}{\partial t}$$

$$\Rightarrow \quad \frac{\partial V}{\partial t} = 4I_2 - 4V + 4i \qquad \text{....(ii)}$$

Arranging eq. (i) and eq. (ii)

$$\frac{\partial V}{\partial t} = -4V + 4i + 4I_2$$

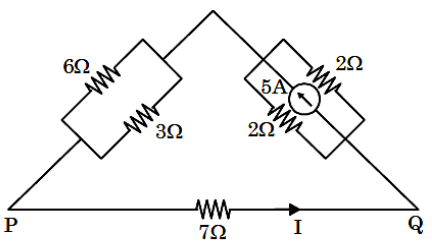
$$\frac{\partial u}{\partial t} = -2V - 4i + 4i_1$$

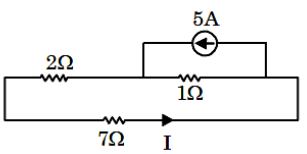
$$\frac{\partial}{\partial t}\begin{bmatrix} V \\ i \end{bmatrix} = \begin{bmatrix} -4 & 4 \\ -2 & -4 \end{bmatrix}\begin{bmatrix} V \\ i \end{bmatrix} + \begin{bmatrix} 0 & 4 \\ 4 & 0 \end{bmatrix}\begin{bmatrix} i_1 \\ i_2 \end{bmatrix}$$

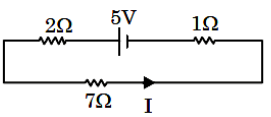
So, option (A) is correct.

2021

1. 0.5 to 0.5



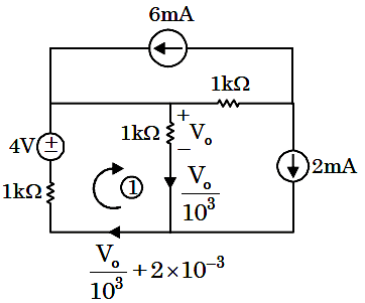




$$\Rightarrow \quad I = \frac{5}{10} = 0.5\text{ A}$$

2. 1.0 to 1.0

kVL is loop (1)



$V_o + V_o - 4 + 2 = 0$

$\Rightarrow \quad 2V_o = 2$

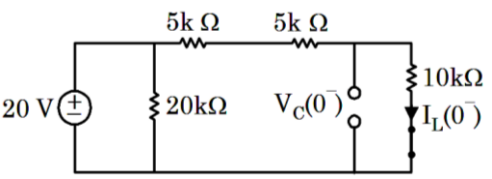
$V_o = 1$ V

3. (C)

At t = 0^- i.e. steady state

Capacitor $\rightarrow$ o/c

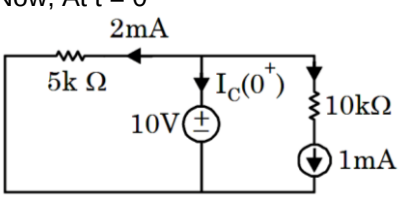
Inductor $\rightarrow$ s/c



$I_L = (0^-) = 1$ mA

$V_C(0^-) = 10$ V

Now, At t = 0^+



$I_C(0^+) + 2 + 1 = 0$

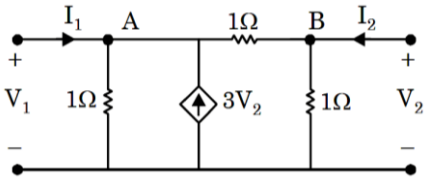
$I_C(0^+) = -3$ mA

$I_C = C\dfrac{dV}{dt}$

$\Rightarrow \quad \dfrac{-3\text{mA}}{1\text{mA}} = \dfrac{dV}{dt}$

$\Rightarrow \quad -3\text{V / sec} = \dfrac{dV}{dt}$

4. (C)



KCL at node A:

$$I_1 = V_1 + \frac{V_1 - V_2}{1} - 3V_2$$

$I_1 = 2V_1 - 4V_2$ ….(i)

$$I_2 = \frac{V_2}{1} + \frac{V_2 - V_1}{1}$$

$I_2 = -V_1 + 2V_2$ ….(ii)

From equation (i) & (ii),

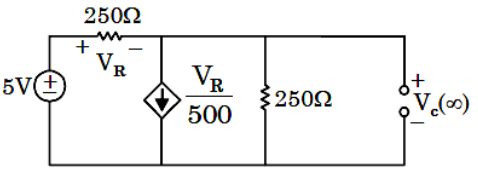
$$[Y] = \begin{bmatrix} 2 & -4 \\ -1 & 2 \end{bmatrix}$$

5. 0.132 to 0.146

Given: $V_c(0^-) = -5V$

$\therefore$ $V_c(0^-) = V_c(0^-) = -5V$

At $t = \infty$



$$\frac{V_c(\infty)}{250} + \frac{V_c(\infty) - 5}{250} + \frac{V_R}{500} = 0$$

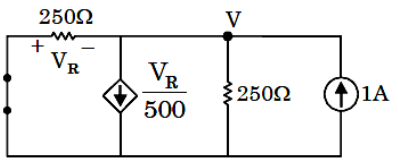
and $-5 + V_R + V_{c(\infty)} = 0$

$$V_R = 5 - V_{c(\infty)}$$

$$\therefore \quad \frac{V_c(\infty)}{250} + \frac{V_c(\infty) - 5}{250} + \frac{5 - V_c(\infty)}{500} = 0$$

$$V_c(\infty) = \frac{5}{3} = 1.66\text{ V}$$

Now for calculation of time constant:



$$\frac{V}{250} + \frac{V}{250} + \frac{V_R}{500} = 1$$

$\because \quad V_R = -V$

$\therefore \quad V = \frac{500}{3} V$

$\therefore \quad R_{eq} = \frac{V}{1} = \frac{500}{3} \Omega$

Time constant = RC = $\frac{500}{3} \times 0.6 \times 10^{-6}$

$\tau = 0.1 \times 10^{-3}$ sec

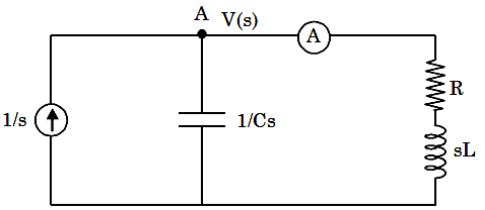
$\therefore \quad V_c(t) = V_c(\infty) + \left[V_c(0^+) - V_c(\infty)\right] e^{-t/\tau}$

$\Rightarrow \quad 0 = \frac{5}{3} + \left[-5 - \frac{5}{3}\right] e^{-10^4 t}$

$\Rightarrow$ t = 0.139 m sec

6. 1.40 to 1.50

Convert into Laplace domain:



Nodal at a :

$$\frac{V(s)}{1/Cs} + \frac{V(s)}{R + sL} = \frac{1}{s}$$

$$\Rightarrow \quad V(s) = \frac{(1/s)[R + sL]}{LCs^2 + RCs + 1}$$

$$\Rightarrow \quad I(s) = \frac{1}{s[LCs^2 RCs + 1]} = \frac{1}{sLC\left[s^2 + \frac{RC}{LC}s + \frac{1}{LC}\right]}$$

$$\xi = \frac{R}{2}\sqrt{C/L} = 0.25$$

Maximum over shoot

$\%M_P = e^{-\frac{\pi\xi}{\sqrt{1-\xi^2}} \times 100}$

$\%M_P = 44$

As $\% M_P = \frac{I(t_p) - I_{(\infty)}}{I_{(\infty)}} \times 100$

$\Rightarrow \quad \frac{I(t_P) - 1}{1} \times 100 = 44$

$I(t_P) = 1.44$

$I(t_P)$ i.e. maximum ammeter reading = 1.44

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